

## DAY THIRTY EIGHT

# Mock Test 1

### Instructions

- The test consists of 45 questions.
- Candidates will be awarded marks for correct response of each question. 1/4 (one-fourth) marks will be deducted for indicating incorrect response of each question. No deduction from the total score will be made, if no response is indicated for an item in the answer sheet.
- There is only one correct response for each question. Filling up more than one response in each question will be treated as wrong response.

- 1 Railway wagon axles are made by heating rods of iron embedded in charcoal powder. The process is known as  
(a) case hardening (b) sheradizing  
(c) annealing (d) tempering
- 2 Equal volumes of monoatomic and diatomic gases at same initial temperature and pressure are mixed. The ratio of specific heats of the mixture ( $C_p / C_v$ ) will be  
(a) 1 (b) 2 (c) 1.67 (d) 1.5
- 3 28 g of  $N_2$  and 6 g of  $H_2$  were kept at  $400^\circ C$  in 1 L vessel, the equilibrium mixture contained 27.54 g of  $NH_3$ . The approximate value of  $K_c$  for the above reaction can be (in  $mol^{-2} L^2$ )  
(a) 75 (b) 50 (c) 25 (d) 100
- 4 Aluminium chloride exists as dimer,  $Al_2Cl_6$  in solid state as well as in solution of non-polar solvents such as benzene. When dissolved in water, it gives  
(a)  $Al^{3+} + 3Cl^-$  (b)  $[Al(H_2O)_6]^{3+} + 3Cl^-$   
(c)  $[Al(OH)_6]^{3-} + 3HCl$  (d)  $Al_2O_3 + 6HCl$
- 5 The dissociation energy of  $CH_4$  and  $C_2H_6$  to convert them into gaseous atoms are 360 and 620  $kcal\ mol^{-1}$  respectively. The bond energy of C—C bond is  
(a) 260  $kcal\ mol^{-1}$  (b) 180  $kcal\ mol^{-1}$   
(c) 130  $kcal\ mol^{-1}$  (d) 80  $kcal\ mol^{-1}$
- 6 When 10 mL of 0.1 M acetic acid ( $pK_a = 5.0$ ) is titrated against 10 mL of 0.1 M ammonia solution ( $pK_b = 5.0$ ), the equivalence point occurs at pH  
(a) 5.0 (b) 6.0 (c) 7.0 (d) 9.0
- 7 Which of the following salts does not get hydrolysed in water?  
(a)  $KClO_4$  (b)  $NH_4Cl$   
(c)  $CH_3COONa$  (d) None of these
- 8 The axial angles in triclinic crystal system are  
(a)  $\alpha = \beta = \gamma = 90^\circ$  (b)  $\alpha = \gamma = 90^\circ, \beta \neq 90^\circ$   
(c)  $\alpha \neq \beta \neq \gamma \neq 90^\circ$  (d)  $\alpha = \beta = \gamma \neq 90^\circ$
- 9 How many unit cells are present in a cube shaped ideal crystal of NaCl of mass 1.00 g  
(Atomic mass Na = 23, Cl = 35.5) ?  
(a)  $1.28 \times 10^{21}$  unit cells  
(b)  $1.71 \times 10^{21}$  unit cells  
(c)  $2.57 \times 10^{21}$  unit cells  
(d)  $5.14 \times 10^{21}$  unit cells
- 10 Rearrange the following (I to IV) in the order of increasing masses and choose answer from (a), (b), (c) and (d) (Atomic mass : N = 14, O = 16, Cu = 63)  
I. 1 molecule of oxygen  
II. 1 atom of nitrogen  
III.  $1 \times 10^{-10}$  g molecular weight of oxygen  
IV.  $1 \times 10^{-10}$  g atomic weight of copper  
(a) II < I < III < IV (b) IV < III < II < I  
(c) II < III < I < IV (d) III < IV < I < II
- 11 A monoprotic acid in 1.00 M solution is 0.01% ionised. The dissociation constant of this acid is  
(a)  $1 \times 10^{-8}$  (b)  $1 \times 10^{-4}$   
(c)  $1 \times 10^{-6}$  (d)  $1 \times 10^{-5}$

12 Observe the following abbreviations

$\pi_{\text{obs}}$  = observed colligative property

$\pi_{\text{cal}}$  = theoretical colligative property assuming normal behaviour of solute

van't Hoff factor ( $i$ ) is given by

- (a)  $i = \pi_{\text{obs}} \times \pi_{\text{cal}}$  (b)  $i = \pi_{\text{obs}} + \pi_{\text{cal}}$   
 (c)  $i = \pi_{\text{obs}} - \pi_{\text{cal}}$  (d)  $i = \frac{\pi_{\text{obs}}}{\pi_{\text{cal}}}$

13 Among the following, the most stable compound is

- (a) *cis*-1, 2-cyclohexanediol  
 (b) *trans*-1, 2-cyclohexanediol  
 (c) *cis*-1, 3-cyclohexanediol  
 (d) *trans*-1, 3-cyclohexanediol

14 Ionic compounds are formed most easily by elements with

- (a) low electron affinity, high ionisation energy  
 (b) high electron affinity, low ionisation energy  
 (c) low electron affinity, low ionisation energy  
 (d) high electron affinity, high ionisation energy

15 The bond order in NO is 2.5 while that in  $\text{NO}^+$  is 3. Which of the following statements is true for these two species?

- (a) Bond length in  $\text{NO}^+$  is equal to that in NO  
 (b) Bond length in NO is greater than in  $\text{NO}^+$   
 (c) Bond length in  $\text{NO}^+$  is greater than in NO  
 (d) Bond length is unpredictable

16 The main product obtained when a solution of sodium carbonate react with mercuric chloride is

- (a)  $\text{Hg}(\text{OH})_2$  (b)  $\text{HgCO}_3 \cdot \text{HgO}$   
 (c)  $\text{HgCO}_3$  (d)  $\text{HgCO}_3 \cdot \text{Hg}(\text{OH})_2$

17 The change in optical rotation with time, of freshly prepared solution of sugar is known as

- (a) rotatory motion (b) inversion  
 (c) specific rotation (d) mutarotation

18 Which is the most acidic of the following?

- (a) Methane (b) Acetylene  
 (c) 1-butene (d) *Neo*-pentane

19 The maximum number of  $90^\circ$  angles between bond pair-bond pair of electrons is observed in following hybridisation

- (a)  $dsp^3$  (b)  $dsp^2$  (c)  $sp^3d$  (d)  $sp^3d^2$

20 A reaction, that is of first order with respect to reactant A, has a rate constant  $6 \text{ min}^{-1}$ . If we start with  $[A] = 0.5 \text{ mol L}^{-1}$ , when would  $[A]$  reach the value  $0.05 \text{ mol L}^{-1}$ ?

- (a) 0.384 min (b) 0.15 min  
 (c) 3 min (d) 3.84 min

21 Which pair of compounds is expected to show similar colour in aqueous medium?

- (a)  $\text{FeCl}_2$  and  $\text{CuCl}_2$  (b)  $\text{VOCl}_2$  and  $\text{CuCl}_2$   
 (c)  $\text{VOCl}_2$  and  $\text{FeCl}_2$  (d)  $\text{FeCl}_2$  and  $\text{MnCl}_2$

22 A vessel at 1000 K contains  $\text{CO}_2$  with a pressure of 0.5 atm. Some of the  $\text{CO}_2$  is converted into CO on the addition of graphite. If the total pressure at equilibrium is 0.8 atm, the value of  $K_p$  is

- (a) 1.8 atm (b) 3 atm (c) 0.3 atm (d) 0.18 atm

23  $\text{CH}_3\text{MgI}$  will give methane with

- (a)  $\text{C}_2\text{H}_5\text{OH}$  (b)  $\text{CH}_3\text{—CH}_2\text{—NH}_2$   
 (c)  $\text{CH}_3\text{—CO—CH}_3$  (d) Both (a) and (b)

24  $E^\circ$  values of  $\text{Mg}^{2+}/\text{Mg}$  is  $-2.37 \text{ V}$ , of  $\text{Zn}^{2+}/\text{Zn}$  is  $-0.76 \text{ V}$  and  $\text{Fe}^{2+}/\text{Fe}$  is  $-0.44 \text{ V}$ . Which of the statements is correct?

- (a) Zn will reduce  $\text{Fe}^{2+}$  (b) Zn will reduce  $\text{Mg}^{2+}$   
 (c) Mg oxidises Fe (d) Zn oxidises Fe

25 A compound A, has a molecular formula  $\text{C}_2\text{Cl}_3\text{OH}$ . It reduces Fehling solution and on oxidation gives a monocarboxylic acid (B). A is also obtained by the action of chlorine on ethyl alcohol. A is

- (a) chloral (b)  $\text{CHCl}_3$   
 (c)  $\text{CH}_3\text{Cl}$  (d) chloroacetic acid

26 At low temperature, phenol reacts with  $\text{Br}_2$  in  $\text{CS}_2$  to form

- (a) *m*-bromophenol (b) *o* and *p*-bromophenol  
 (c) *p*-bromophenol (d) 2, 4, 6-tribromophenol

27 An organic compound on analysis gave C = 48%, H = 8% and N = 56%. Volume of 1.0 g of the compound was found to be 200 mL at NTP. Molecular formula of the compound is

- (a)  $\text{C}_4\text{H}_8\text{N}_4$  (b)  $\text{C}_2\text{H}_4\text{N}_2$   
 (c)  $\text{C}_{12}\text{H}_{24}\text{N}_{12}$  (d)  $\text{C}_{16}\text{H}_{32}\text{N}_{16}$

28 General configuration of outermost and penultimate shell is  $(n-1)s^2p^6(n-1)d^xns^2$ . If  $n=4$  and  $x=5$ , number of protons in the nucleus will be

- (a) < 24 (b) 30 (c) 25 (d) > 25

29 Which property of  $\text{Na}_2\text{S}_2\text{O}_3$  makes it useful in photography?

- (a) Photochemical property  
 (b) Complex formation property  
 (c) Oxidising agent  
 (d) Reducing agent

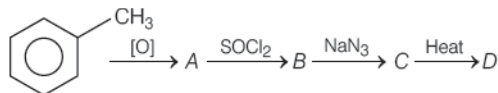
30 A gas decolourised  $\text{KMnO}_4$  solution but gives no precipitate with ammoniacal cuprous chloride. Which of the following gases does not give a precipitate with ammoniacal solution of silver nitrate but decolourises  $\text{KMnO}_4$  (neutral or slightly alkaline)?

- (a) Ethane (b) Methane  
 (c) Ethene (d) Acetylene

31 Which one of the following undergoes reaction with 50% sodium hydroxide solution to give the corresponding alcohol and acid salt?

- (a) Butanol (b) Benzaldehyde  
 (c) Phenol (d) Benzoic acid

32 In the following sequence of reactions, what is D ?



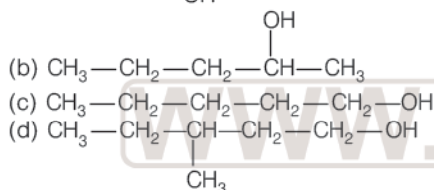
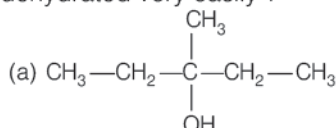
- (a) Primary amine  
(b) An amide  
(c) Phenyl isocyanate  
(d) A chain lengthend hydrocarbon

33  $2C + O_2 \longrightarrow 2CO$ ;  $\Delta H = -220 \text{ kJ}$

Which of the following statements is correct for this reaction?

- (a) Heat of combustion of carbon is 110 kJ  
(b) Reaction is exothermic  
(c) Reaction needs no initiation  
(d) All of the above are correct

34 Among the following compounds, which can be dehydrated very easily ?



35 Acid catalysed hydration of alkenes except ethene leads to the formation of

- (a) mixture of secondary and tertiary alcohols  
(b) mixture of primary and secondary alcohols  
(c) secondary or tertiary alcohol  
(d) primary alcohol

36 What is obtained when chlorine is passed in boiling toluene and product is hydrolysed?

- (a) *o*-cresol (b) *p*-cresol  
(c) 2, 4-dihydroxy toluene (d) Benzyl alcohol

37  $\text{CaC}_2 + \text{H}_2\text{O} \longrightarrow A \xrightarrow{\text{H}_2\text{SO}_4/\text{HgSO}_4} B$

Identify A and B in the given reaction.

- (a)  $\text{C}_2\text{H}_2$  and  $\text{CH}_3\text{CHO}$  (b)  $\text{C}_2\text{H}_4$  and  $\text{CH}_3\text{COOH}$   
(c)  $\text{CH}_4$  and  $\text{HCOOH}$  (d)  $\text{C}_2\text{H}_2$  and  $\text{CH}_3\text{COOH}$

38 Which of the following is not a green house gas?

- (a)  $\text{CO}_2$  (b) Chlorofluorocarbons  
(c)  $\text{CH}_4$  (d) None of these

39 The saponification value of an oil or fat is measured in terms of

- (a)  $\text{NH}_4\text{OH}$  (b)  $\text{NaOH}$  (c)  $\text{KOH}$  (d)  $\text{C}_6\text{H}_5\text{OH}$

40 An aqueous solution of  $\text{CoCl}_2$  on addition of excess of concentrated  $\text{HCl}$  turns blue due to the formation of

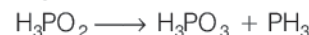
- (a)  $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]$  (b)  $[\text{Co}(\text{H}_2\text{O})_2\text{Cl}_4]^{2-}$   
(c)  $[\text{CoCl}_4]^{2-}$  (d)  $[\text{Co}(\text{H}_2\text{O})_2\text{Cl}_2]$

**Direction** (Q. Nos. 41-42) In the following questions more than one of the answers given may be correct. Select the correct answers and mark it according to the codes.

**Codes**

- (a) 1, 2 and 3 are correct (b) 1 and 2 are correct  
(c) 2 and 4 are correct (d) 1 and 3 are correct

41 In the following reaction



1.  $\text{H}_3\text{PO}_2$  undergoes disproportionation reaction  
2. equivalent weight of  $\text{H}_3\text{PO}_2$  is 49.5  
3. equivalent weight of  $\text{H}_3\text{PO}_2$  is 22  
4.  $\text{NaH}_2\text{PO}_2$  is an acid salt

42  $\text{RCONH}_2$  reacts with a mixture of  $\text{Br}_2$  and  $\text{KOH}$  to yield  $\text{R—NH}_2$  as a main product. The intermediates involved in this reaction are

1.  $\text{RNHBr}$  2.  $\text{RCONHBr}$   
3.  $\text{RCONBr}_2$  4.  $\text{R—N}=\text{C}=\text{O}$

43 Match the salt (in Column I) with its relation for  $K_{sp}$  (in Column II) and choose the correct code given below.

| Column I               |    | Column II |
|------------------------|----|-----------|
| A. Aluminium phosphate | 1. | $4s^3$    |
| B. Calcium chloride    | 2. | $27s^4$   |
| C. Ferric sulphate     | 3. | $s^2$     |
| D. Chromium hydroxide  | 4. | $108s^5$  |

**Codes**

- |       |   |   |   |       |   |   |   |
|-------|---|---|---|-------|---|---|---|
| A     | B | C | D | A     | B | C | D |
| (a) 3 | 1 | 4 | 2 | (b) 2 | 3 | 1 | 4 |
| (c) 3 | 2 | 4 | 1 | (d) 4 | 1 | 2 | 3 |

**Direction** (Q. Nos. 44-45) Each of these questions contains two statements : Assertion and Reason. Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select one of the codes (a), (b), (c) and (d) given below.

- (a) Assertion is true, Reason is true; Reason is the correct explanation for Assertion  
(b) Assertion is true, Reason is true; Reason is not the correct explanation for Assertion  
(c) Assertion is true, Reason is false  
(d) Assertion is false, Reason is false

44 **Assertion** A spectral line will be seen for a  $2p_x - 2p_y$  transition.

**Reason** Energy is released in the form of wave of light when the electron drops from  $2p_x$  to  $2p_y$  orbital.

45 **Assertion** Amongst the halogen, fluorine can oxidise the elements to high oxidation state.

**Reason** Due to small size of fluoride ion, it is difficult to oxidise fluoride ion to fluorine. Hence, reverse reaction takes place more easily.

## ANSWERS

|        |        |        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 1 (a)  | 2 (d)  | 3 (a)  | 4 (b)  | 5 (d)  | 6 (c)  | 7 (a)  | 8 (c)  | 9 (c)  | 10 (a) |
| 11 (a) | 12 (d) | 13 (d) | 14 (b) | 15 (b) | 16 (b) | 17 (d) | 18 (b) | 19 (d) | 20 (a) |
| 21 (b) | 22 (a) | 23 (d) | 24 (a) | 25 (a) | 26 (b) | 27 (a) | 28 (c) | 29 (b) | 30 (c) |
| 31 (b) | 32 (c) | 33 (b) | 34 (a) | 35 (c) | 36 (d) | 37 (a) | 38 (d) | 39 (c) | 40 (c) |
| 41 (b) | 42 (c) | 43 (a) | 44 (d) | 45 (c) |        |        |        |        |        |

## Hints and Explanations

**1** Case hardening is the process of hardening of the surface of wrought iron by depositing a surface layer of steel on it. It is done by heating wrought iron in contact with potassium ferrocyanide. Alternatively, case hardening can also be done by heating wrought iron with charcoal and then, plunging it in a suitable oil.

$$2 \quad C_V = \frac{3}{2}RT; \quad C_P = \frac{5}{2}RT$$

(for monoatomic gas)

$$C_V = \frac{5}{2}RT; \quad C_P = \frac{7}{2}RT \quad (\text{for diatomic gas})$$

Thus, for mixture of 1 mole each,

$$C_V = \frac{\frac{3}{2}RT + \frac{5}{2}RT}{2}$$

$$\text{and } C_P = \frac{\frac{5}{2}RT + \frac{7}{2}RT}{2}$$

$$\text{Therefore, } \frac{C_P}{C_V} = \frac{3RT}{2RT} = 1.5$$

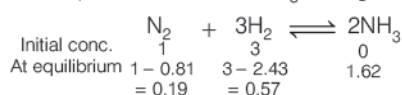
$$3 \quad \text{Number of moles of } N_2 = \frac{28}{28} = 1 \text{ mol}$$

$$\text{Number of moles of } H_2 = \frac{6}{2} = 3 \text{ mol}$$

Number of moles of

$$NH_3 = \frac{27.54}{17} = 1.62 \text{ mol}$$

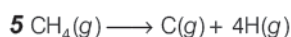
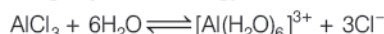
( $\therefore$  Molecular wt. of  $NH_3 = 17 \text{ g/mol}$ )



$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

$$= \frac{[1.62]^2}{[0.19][0.57]^3} = 75$$

**4**  $AlCl_3$  is covalent and exists as dimer i.e.  $Al_2Cl_6$  but in water, it becomes ionic due to large hydration energy of  $Al^{3+}$



$\Delta H = 360 \text{ kcal}$  for four (C—H) bonds

$$BE(C-H) = \frac{360}{4} = 90 \text{ kcal}$$



$\Delta H = 620 \text{ kcal}$  for six (C—H) bonds and one (C—C) bond

$$\therefore 6 BE(C-H) + BE(C-C) = 620$$

$$6 \times 90 + BE(C-C) = 620$$

$$BE(C-C) = 80 \text{ kcal mol}^{-1}$$

$$6 \quad pK_a = -\log K_a, \quad pK_b = -\log K_b$$

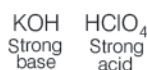
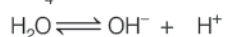
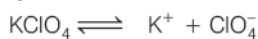
$$pH = -\frac{1}{2}[\log K_a + \log K_w - \log K_b]$$

$$= -\frac{1}{2}[-5 + \log(1 \times 10^{-14}) - (-5)]$$

$$[\because K_w = 1 \times 10^{-14}]$$

$$= -\frac{1}{2}[-5 - 14 + 5] = -\frac{1}{2}[-14] = 7$$

**7** Only salts of weak acid + strong base and strong acid + weak base get hydrolysed (i.e. show alkalinity or acidity in water).  $KClO_4$  is a salt of strong acid and strong base, therefore it does not get hydrolysed in water.



**8** The axial angles in triclinic crystal system are different and none is perpendicular to any of the others, i.e.

$$\alpha \neq \beta \neq \gamma \neq 90^\circ$$

( $\alpha = \beta = \gamma = 90^\circ$  for orthorhombic)

( $\alpha = \gamma = 90^\circ; \beta \neq 90^\circ$  for monoclinic)  
( $\alpha = \beta = \gamma \neq 90^\circ$  for rhombohedral)

**9** Number of NaCl in 1 g

$$= \frac{1}{58.5} \times 6.023 \times 10^{23}$$

$$= 1.029 \times 10^{22}$$

A unit cell contains  $4Na^+$  and  $4Cl^-$  ions.

$\therefore$  Number of unit cells in a NaCl crystal of 1g

$$= \frac{1.029 \times 10^{22}}{4}$$

$$= 2.57 \times 10^{21} \text{ unit cells.}$$

**10** I. For 1 molecule of oxygen

$\therefore 6.023 \times 10^{23}$  molecule has mass

$$= 32 \text{ g}$$

$\therefore$  1 molecule of  $O_2$  has mass

$$= \frac{32}{6 \times 10^{23}}$$

$$= 5.3 \times 10^{-23} \text{ g}$$

II. For 1 molecule of nitrogen

$\therefore 2 \times 6 \times 10^{23}$  atoms of  $N_2$  has mass

$$= 28 \text{ g}$$

$\therefore$  1 atom of  $N_2$  has mass

$$= \frac{28}{2 \times 6 \times 10^{23}} = 2.3 \times 10^{-23} \text{ g}$$

III.  $1 \times 10^{-10} \text{ g}$  molecular weight of oxygen

Gram atomic weight

$$= 2 \times 1 \times 10^{-10}$$

$$= 2 \times 10^{-10} \text{ g}$$

IV. Atomic weight of copper  $1 \times 10^{-10} \text{ g}$

So, order of increasing mass is

$$II < I < III < IV.$$

- 11 For weak electrolytes, according to Ostwald's dilution law

$$\alpha = \sqrt{KV}$$

Here,  $\alpha = 0.01\% = 0.0001 = 1 \times 10^{-4}$

$$V = \frac{1}{C} = \frac{1}{1.0} = 1 \text{ L}$$

$$\therefore K_a = \frac{\alpha^2}{V} = \frac{(1 \times 10^{-4})^2}{1} = 1 \times 10^{-8}$$

- 12 van't Hoff factor ( $i$ ) is given by

$$i = \frac{\text{Observed value of colligative property}}{\text{Normal value of colligative property}}$$

The normal value of colligative property is the theoretically calculated value assuming no association or dissociation.

$$\therefore i = \frac{\pi_{\text{obs}}}{\pi_{\text{cal}}}$$

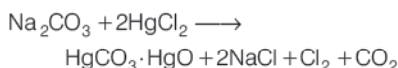
- 13 *Trans* configuration is more stable than *cis* configuration because in *cis* configuration, the higher groups (high molecular weight) are thrown closely enough together to cause crowding or repulsion. Again between 1, 2 and 1, 3-configuration, in 1, 3-configuration the —OH groups are placed farther apart to minimise the repulsion. Hence, more stable is 1, 3-configuration.

- 14 The formation of ionic bond depends upon ease of formation of cation and anion. Therefore, the ionisation energy value of the metal atom should be low, so that it can easily form cation. On the other hand, the electron affinity value of the non-metal atom should be high, so that it can easily form anion.

- 15 Bond length  $\propto \frac{1}{\text{bond order}}$

Higher the bond order, shorter will be the bond length, thus NO (having lower bond order, i.e. 2.5) as compared to NO<sup>+</sup> (having higher bond order i.e. 3) has greater bond length.

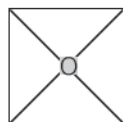
- 16 Basic mercuric carbonate is obtained in this reaction



- 17  $\alpha$ -D-glucose  $\rightleftharpoons$  Equilibrium mixture  
 $[\alpha]_D = +112^\circ$   $[\alpha]_D = +52^\circ$   
 (36%) (0.02%)  
 $\rightleftharpoons$   $\beta$ -D-glucose  
 $[\alpha]_D = +19^\circ$   
 (64%)

Glucose has two forms  $\alpha$ - and  $\beta$ - form. When either of these two is dissolved in water and allowed to stand, it gets converted into an equilibrium mixture of  $\alpha$ - and  $\beta$ - forms with different specific rotation. This change in specific rotation in aqueous solution is called **mutarotation**.

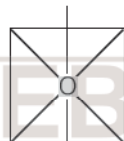
- 18 The  $s$ -character of C—H bond of acetylene is higher in comparison to C—H bond of ethene and ethane. The electrons of the C—H bond in acetylene are strongly held by carbon nuclei. This facilitates the removal of hydrogen as proton.



$dsp^2$ -hybridisation  
(four  $90^\circ$  angles  
between  $bp$ - $bp$ )



$sp^3d$  or  $dsp^3$ -hybridisation  
(six  $90^\circ$  angles  
between  $bp$ - $bp$ )



$sp^3d^2$ -hybridisation  
(twelve  $90^\circ$  angles  
between  $bp$ - $bp$ )

$sp^3d^2$  hybridisation has octahedral or square bipyramidal configuration. All the bond angles are in  $90^\circ$  in it.

- 20 We know that for first order kinetics,

$$k = \frac{2.303}{t} \log \frac{a}{a-x}$$

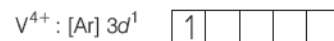
$$a = 0.5 \text{ mol L}^{-1}$$

$$(a-x) = 0.05 \text{ mol L}^{-1}$$

$$6 = \frac{2.303}{t} \log \frac{0.5}{0.05}$$

$$\text{or } t = \frac{2.303}{6} \log \frac{0.5}{0.05} = \frac{2.303}{6} = 0.384 \text{ min}$$

- 21 Colour of transition metal ion salt is due to  $d$ - $d$  transition of unpaired electrons of  $d$ -orbital. Metal ion salts having similar number of unpaired electrons in  $d$ -orbital show similar colour in aqueous medium.



Number of unpaired electrons = 1

So both will show same colour.

- 22  $\text{CO}_2(g) + \text{C}(s) \rightleftharpoons 2\text{CO}(g)$

Initial : 0.5 atm

At equilibrium :  $(0.5-p)$   $2p$  atm

This is a case of heterogeneous equilibrium. C(s) being solid is not considered.

Total pressure of  $\text{CO}_2$  and CO gases.

$$P_{\text{CO}_2} + P_{\text{CO}} = P_{\text{total}}$$

$$0.5 - p + 2p = 0.8,$$

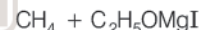
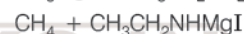
$$\therefore P_{\text{CO}_2} = 0.5 - 0.3 = 0.2 \text{ atm}$$

$$[\because p = 0.3 \text{ atm}]$$

$$P_{\text{CO}} = 2p = 0.6 \text{ atm}$$

$$K_p = \frac{P_{\text{CO}}^2}{P_{\text{CO}_2}} = \frac{0.6 \times 0.6}{0.2} = 1.8 \text{ atm}$$

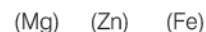
- 23  $\text{CH}_3\text{MgI} + \text{CH}_3\text{CH}_2\text{NH}_2 \longrightarrow$



Here, alkyl group of Grignard's reagent is involved in the formation of alkane.

- 24 Higher the negative value of  $E^\circ$ , more is the reducing power. The order of  $E^\circ$  values (negative value) is

$$-2.37 > -0.76 > -0.44$$



$\therefore$  Mg can reduce both  $\text{Zn}^{2+}$  and  $\text{Fe}^{2+}$ . Zn can reduce  $\text{Fe}^{2+}$ , but not  $\text{Mg}^{2+}$ . Fe cannot reduce Mg and Zn but can oxidise them.

- 25  $\text{C}_2\text{Cl}_3\text{OH} + \text{Fehling solution} \longrightarrow$



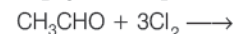
Red precipitate

It means —CHO group is present

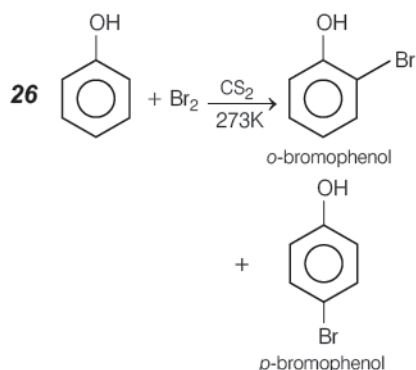


Monocarboxylic acid

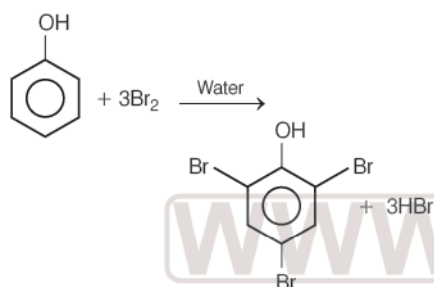
It means only one —CHO group is present.



Chloral



In the presence of non-polar solvent ( $\text{CS}_2$ ), the ionisation of phenol is suppressed. The ring is slightly activated and hence, mono substitution occurs. On the other hand, with  $\text{Br}_2$  water phenol forms 2, 4, 6-tribromophenol.



In aqueous solution, phenol ionises to give phenoxide ion. Due to the presence of negative charge on oxygen the benzene ring is highly activated and hence, trisubstituted product is obtained.

27

| Element | Atomic number | Percent age | Number of moles     | Simple ratio |
|---------|---------------|-------------|---------------------|--------------|
| C       | 12            | 48          | $\frac{48}{12} = 4$ | 1            |
| H       | 1             | 8           | $\frac{8}{1} = 8$   | 2            |
| N       | 14            | 56          | $\frac{56}{14} = 4$ | 1            |

Empirical formula =  $\text{CH}_2\text{N}$

Empirical formula mass = 28

Now, 200 mL of compound = 1 g

$\therefore$  22400 mL of compound

$$= \frac{1}{200} \times 22400 = 112 \text{ g}$$

= molecular mass

$$n = \frac{\text{molecular mass}}{\text{empirical formula mass}} = \frac{112}{28} = 4$$

Therefore, molecular formula

$$= (\text{CH}_2\text{N})_4 = \text{C}_4\text{H}_8\text{N}_4$$

28 Given,  $n = 4$ ,  $x = 5$

$$(4-1)s^2 (4-1)p^6 (4-1)d^5 4s^2$$

$$3s^2 \quad 3p^6 \quad 3d^5 \quad 4s^2$$

Total electrons =  $2 + 6 + 5 + 2 = 15$

Electrons in I and II orbits =  $2 + 8 = 10$

$\therefore$  Total electrons =  $15 + 10 = 25$

Number of electrons = number of protons

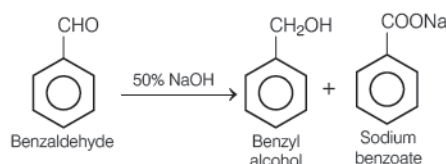
$\therefore$  Total protons = 25

29 Sodium thiosulphate ( $\text{Na}_2\text{S}_2\text{O}_3$ ) is useful in photography due to its complex formation property. It is used in photography as a fixer since, it dissolves unexposed silver bromide.

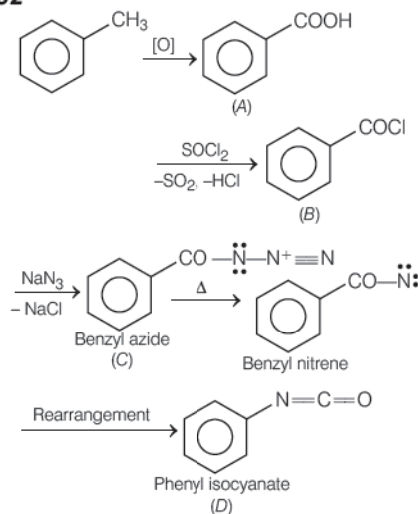
30 Ethane and methane do not decolourise  $\text{KMnO}_4$  and do not react with ammoniacal cuprous chloride. Acetylene decolourises  $\text{KMnO}_4$  solution and also gives red precipitate with ammoniacal cuprous chloride.

On the other hand, ethene decolourises  $\text{KMnO}_4$  solution but does not react with ammoniacal cuprous chloride.

31 Benzaldehyde will undergo Cannizzaro's reaction on treatment with 50%  $\text{NaOH}$  to produce benzyl alcohol and benzoic acid as it does not contain  $\alpha$ -hydrogen.



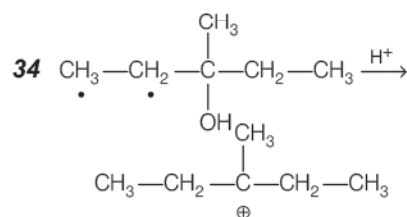
32



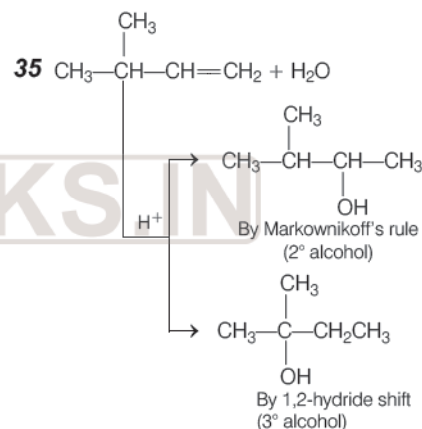
33  $2\text{C} + \text{O}_2 \longrightarrow 2\text{CO}$ ;  $\Delta H = -220 \text{ kJ}$

This reaction does not represent complete combustion of carbon, hence heat of combustion of carbon will not be equal to 110 kJ.

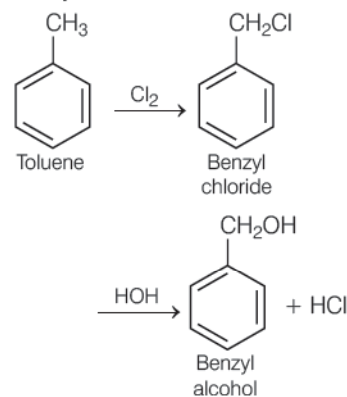
The negative sign of  $\Delta H$  indicates that this reaction is exothermic. Also, despite being spontaneous reaction, it requires initiation.



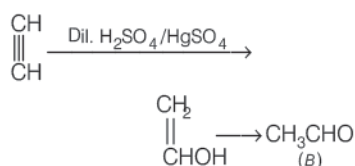
The more stable carbocation is generated, thus more easily it will be dehydrated.



36 When chlorine is passed in boiling toluene (free radical) substitution in side chain takes place and benzyl chloride is obtained which on hydrolysis gives benzyl alcohol.



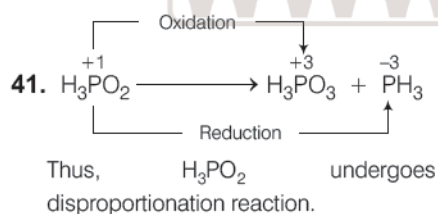
37  $\text{CaC}_2 + 2\text{H}_2\text{O} \longrightarrow \text{C}_2\text{H}_2 + \text{Ca(OH)}_2$   
(A)



**38** Carbon dioxide, methane, nitrous oxide, ozone, chlorofluorocarbons, water vapour and carbon tetrachloride are the principal greenhouse gases. If the atmospheric concentration of these absorbing gases increases, more heat will be trapped in the atmosphere which will become warmer and overall increase in temperature would be observed (global warming).

**39** Saponification value is the number of milligrams of KOH required to neutralise the fatty acid resulting from the complete hydrolysis of 1 g of oil or fat.

**40**  $\text{CoCl}_2$  is a weak Lewis acid, so reacts with chloride ion to produce salt containing the tetrahedral  $[\text{CoCl}_4]^{2-}$  ion.  $\text{CoCl}_2$  is blue when anhydrous, and deep magenta colour when hydrated, for this reason it is widely used as an indicator for water.

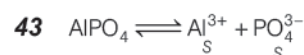
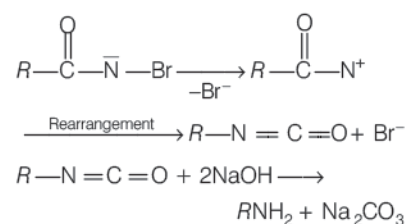
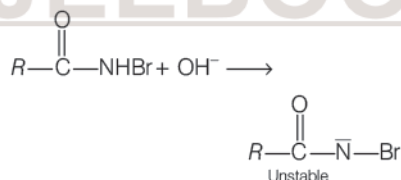
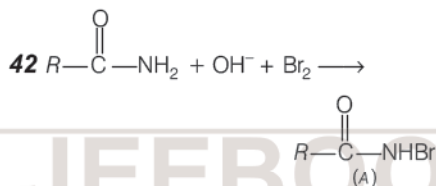
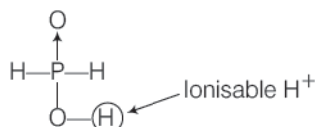


$$E_{(\text{ox})} = \frac{\text{molecular weight}}{\text{increase in oxidation number}} = \frac{66}{2} = 33$$

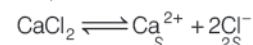
$$E_{(\text{red})} = \frac{\text{molecular weight}}{\text{decrease in oxidation number}} = \frac{66}{4} = 16.5$$

$$\therefore E_{\text{net}} = E_{(\text{ox})} + E_{(\text{red})} = 33 + 16.5 = 49.5$$

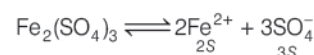
$\text{H}_3\text{PO}_2$  is a monobasic salt with only one ionisable  $\text{H}^+$ . Hence,  $\text{NaH}_2\text{PO}_2$  is not acid salt.



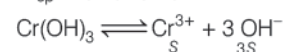
$$\therefore K_{\text{sp}} = S^2$$



$$\therefore K_{\text{sp}} = S(2S)^2 = 4S^3$$



$$\therefore K_{\text{sp}} = (2S)^2(3S)^3 = 108S^5$$



$$K_{\text{sp}} = S(3S)^3 = 27S^4$$

**44** Both the Assertion and Reason are false because a spectral line is observed when an electron comes back from the excited state to the ground state in one or more jump. In each jump it come back from higher energy level to lower energy level corresponding to  $n = 1, 2, 3, 4, \dots$  called  $K, L, M, N \dots$  shells. So, shell must be change.

**45** Assertion is true but Reason is false because fluorine can oxidise the other elements to higher oxidation state because of being highly electronegative.

## DAY THIRTY NINE

# Mock Test 2

### Instructions

- The test consists of 45 questions.
- Candidates will be awarded marks for correct response of each question. 1/4 (one-fourth) marks will be deducted for indicating incorrect response of each question. No deduction from the total score will be made, if no response is indicated for an item in the answer sheet.
- There is only one correct response for each question. Filling up more than one response in each question will be treated as wrong response.

- 1 Which of the following is the strongest Lewis acid?  
(a)  $\text{BI}_3$  (b)  $\text{BBr}_3$  (c)  $\text{BCl}_3$  (d)  $\text{BF}_3$
- 2 The correct order of acidic strength of the following compounds is  
(a)  $\text{Cl}_2\text{O}_7 > \text{SO}_2 > \text{P}_4\text{O}_{10}$  (b)  $\text{K}_2\text{O} > \text{CaO} > \text{MgO}$   
(c)  $\text{CO}_2 > \text{N}_2\text{O}_5 > \text{SO}_3$  (d)  $\text{Na}_2\text{O} > \text{MgO} > \text{Al}_2\text{O}_3$
- 3 Which of the following iron salts exists as a dimer?  
(a) Ferric chloride (b) Ferrous chloride  
(c) Ferrous sulphate (d) Mohr's salt
- 4 At high pressure, the compressibility factor  $Z$  is equal to  
(a) unity (b)  $1 - \frac{pb}{RT}$   
(c)  $1 + \frac{pb}{RT}$  (d) zero
- 5 In which of the following group, all do not have  $sp^3d$  hybridisation?  
(a)  $\text{ClF}_3, \text{IF}_3, \text{XeF}_3^+$  (b)  $\text{ICl}_2^-, \text{ClF}_2^-, \text{I}_3^-$   
(c)  $\text{ClF}_3, \text{BrF}_3, \text{IF}_3$  (d)  $\text{PCl}_3, \text{AsCl}_3, \text{PF}_5$
- 6 Nascent hydrogen consists of  
(a) hydrogen atoms with excess of energy  
(b) hydrogen molecules with excess of energy  
(c) hydrogen ions in excited state  
(d) solvated protons
- 7 In which of the following compounds, iron has the lowest oxidation number?  
(a)  $\text{Fe}(\text{CO})_5$  (b)  $\text{Fe}_2\text{O}_3$   
(c)  $\text{K}_4[\text{Fe}(\text{CN})_6]$  (d)  $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$
- 8 The solubility of silver bromide in hypo solution is due to the formation of  
(a)  $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$  (b)  $\text{Ag}_2\text{SO}_3$   
(c)  $[\text{Ag}(\text{S}_2\text{O}_3)]^-$  (d)  $\text{Ag}_2\text{S}_2\text{O}_3$
- 9  $\alpha$ - and  $\beta$ -glucose differ in the orientation of  $-\text{OH}$  group around  
(a)  $\text{C}_1$  (b)  $\text{C}_2$   
(c)  $\text{C}_3$  (d)  $\text{C}_4$
- 10 A common metal that is used for the extraction of some metals from their oxides is  
(a) Cr (b) Fe  
(c) Mn (d) Al
- 11 Which one of the following pairs of elements is called 'chemical twins' because of their very similar chemical properties?  
(a) Mn and W (b) Mo and Tc  
(c) Fe and Re (d) Hf and Zr
- 12 Which among the following will show maximum osmotic pressure?  
(a) 1 M NaCl (b) 1 M  $\text{MgCl}_2$   
(c) 1 M  $(\text{NH}_4)_3\text{PO}_4$  (d) 1 M  $\text{Na}_2\text{SO}_4$
- 13 The ratio of coefficients of  $\text{HNO}_3$ ,  $\text{Fe}(\text{NO}_3)_2$  and  $\text{NH}_4\text{NO}_3$  in the following redox reaction, in the balanced form, will be  
$$\text{Fe} + \text{HNO}_3 \longrightarrow \text{Fe}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3 + \text{H}_2\text{O}$$
  
(a) 10 : 1 : 4 (b) 10 : 4 : 1  
(c) 4 : 10 : 1 (d) 4 : 1 : 10

- 14 Enthalpy of sublimation of a substance is equal to  
→ [NCERT Exemplar]

(a) enthalpy of fusion + enthalpy of vaporisation  
(b) enthalpy of fusion  
(c) enthalpy of vaporisation  
(d) twice the enthalpy of vaporisation

- 15 A dextrorotatory optically active alkyl halide undergoes hydrolysis by  $S_N2$  mechanism. The resulting alcohol is

(a) dextrorotatory  
(b) laevorotatory  
(c) optically inactive due to racemisation  
(d) may be dextro or laevorotatory

- 16 One gram of a monobasic acid when dissolved in 100 g of water lowers the freezing point by  $0.186^\circ\text{C}$ . Now, 0.25 g of the same acid is dissolved and titrated with 15.1 mL of  $N/10$  alkali. The degree of dissociation of the acid is ( $K_f$  for  $\text{H}_2\text{O} = 1.86$ )

(a) 1.66 (b) 0.66  
(c) 0.11 (d) 0.060

- 17 The oxide which is the anhydride of orthophosphoric acid, is

(a)  $\text{P}_4\text{O}_{10}$  (b)  $\text{P}_2\text{O}_5$   
(c)  $\text{P}_4\text{O}_6$  (d)  $\text{P}_2\text{O}_3$

- 18  $[\text{Ag}^+]$  in saturated  $\text{AgCl}$  in the presence of 1 M  $\text{KCl}$  [ $K_{sp}(\text{AgCl}) = 1 \times 10^{-10}$ ] is

(a)  $1 \times 10^{-5}$  M (b)  $1 \times 10^{-20}$  M  
(c)  $1 \times 10^{-10}$  M (d)  $2 \times 10^{-10}$  M

- 19 The pair of compounds having metals in their highest oxidation state is

(a)  $\text{MnO}_2$ ,  $\text{FeCl}_3$  (b)  $[\text{MnO}_4]^-$ ,  $\text{CrO}_2\text{Cl}_2$   
(c)  $[\text{Fe}(\text{CN})_6]^{3-}$ ,  $[\text{Co}(\text{CN})_3]$  (d)  $[\text{NiCl}_4]^{2-}$ ,  $[\text{CoCl}_4]^-$

- 20 Which of the following compound will not undergo Cannizzaro reaction on heating with an alkali solution?

(a)  $\text{CCl}_3\text{CHO}$  (b)  $(\text{CH}_3)_3\text{CCHO}$   
(c)  $\text{HCHO}$  (d)  $\text{C}_6\text{H}_5\text{CHO}$

- 21 Among the following compounds, the most susceptible to nucleophilic attack at the carbonyl group, is

(a)  $\text{MeCOCl}$  (b)  $\text{MeCHO}$   
(c)  $\text{MeCOOMe}$  (d)  $\text{MeCOOCOMe}$

- 22 For the reaction,  $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ , the standard free energy is  $\Delta G^\circ > 0$ . The equilibrium constant ( $K$ ) would be  
→ [NCERT Exemplar]

(a)  $K = 0$  (b)  $K > 1$   
(c)  $K = 1$  (d)  $K < 1$

- 23 If 5.85 g of  $\text{NaCl}$  is dissolved in 90 g of water, the mole fraction of  $\text{NaCl}$  is

(a) 0.1 (b) 0.01  
(c) 0.2 (d) 0.0196

- 24 Which sequence correctly describes the relative bond strength of oxygen molecule, superoxide ion, peroxide ion and unipositive oxygen molecule?

(a)  $\text{O}_2^+ > \text{O}_2^- > \text{O}_2^{2-} > \text{O}_2$  (b)  $\text{O}_2^+ > \text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$   
(c)  $\text{O}_2^{2-} > \text{O}_2^- > \text{O}_2 > \text{O}_2^+$  (d)  $\text{O}_2^{2-} > \text{O}_2^- > \text{O}_2^+ > \text{O}_2$

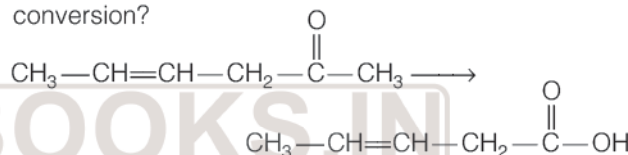
- 25 When an ideal gas undergoes unrestrained expansion, no cooling occurs because the molecules

(a) are above the inversion temperature  
(b) exert no attractive force on each other  
(c) do work equal to loss in kinetic energy  
(d) collide without loss of energy

- 26 A gas is heated in such a way so that its pressure and volume both become double. Again, by lowering the temperature, one fourth of initial number of moles of air has been taken in, to maintain the double volume and pressure. By what fraction, the temperature must have been raised finally?

(a)  $\frac{1}{5}$  time (b)  $\frac{4}{5}$  time (c)  $\frac{16}{5}$  times (d)  $\frac{8}{5}$  times

- 27 Which is the most suitable reagent for the following conversion?



→ [NCERT Exemplar]

(a) Tollen's reagent (b) Benzoyl peroxide  
(c)  $\text{I}_2$  and  $\text{NaOH}$  solution (d)  $\text{Sn}$  and  $\text{NaOH}$  solution

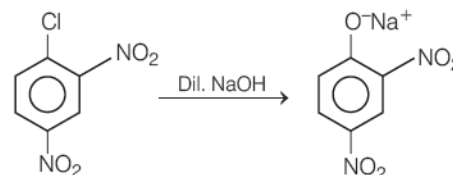
- 28 Nitroethane on treatment with excess of  $\text{Cl}_2$  in the presence of  $\text{NaOH}$  gives

(a)  $\text{ClCH}_2\text{CH}_2\text{NO}_2$  (b)  $\text{CH}_3\text{CHClNO}_2$   
(c)  $\text{Cl}_2\text{CHCH}_2\text{NO}_2$  (d)  $\text{CH}_3\text{CCl}_2\text{NO}_2$

- 29 Temperature dependent equation can be written as

(a)  $\ln k = \ln A - e^{-E_a/RT}$  (b)  $\ln k = \ln A + e^{E_a/RT}$   
(c)  $\ln k = \ln A - e^{RT/E_a}$  (d) None of these

- 30 The following transformation proceeds through



(a) electrophilic addition  
(b) benzyne intermediate  
(c) activated nucleophilic substitution  
(d) oxirane

- 31 A sample of  $\text{CHCl}_3$  before being used as an anaesthetic agent is tested by

(a) Fehling's solution  
(b) ammoniacal cuprous chloride  
(c) silver nitrate solution in cold  
(d) silver nitrate solution after boiling with alcoholic  $\text{KOH}$

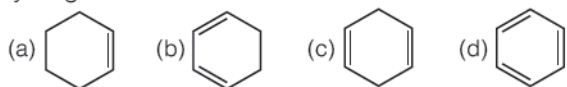
32 When benzaldehyde reacts with acetophenone in the presence of sodium hydroxide, then product is

- (a)  $C_6H_5CH=CHCOC_6H_5$  (b)  $C_6H_5COCH_2C_6H_5$   
(c)  $C_6H_5CH=CHC_6H_5$  (d)  $C_6H_5CH(OH)COC_6H_5$

33 An alkene on vigorous oxidation with  $KMnO_4$  gives only acetic acid. The alkene is

- (a)  $CH_3CH_2CH=CH_2$  (b)  $CH_3CH=CHCH_3$   
(c)  $(CH_3)_2C=CH_2$  (d)  $CH_3CH=CH_2$

34 Which of the following has the maximum heat of hydrogenation?



35 Phenol reacts with  $PCl_5$  to give mainly

- (a) *p*-chlorophenol (b) *o*- and *p*-chlorophenol  
(c) chlorobenzene (d) triphenyl phosphate

36 An aromatic compound X with molecular formula  $C_8H_{10}$  produces one mononitro derivative on nitration and three dinitro derivatives. Compound X would be

- (a) ethyl benzene (b) *m*-xylene  
(c) *o*-xylene (d) *p*-xylene

37 The end product C in the following sequence of chemical reactions, is



- (a) acetaldehyde oxime (b) formaldehyde oxime  
(c) methyl nitrate (d) acetoxime

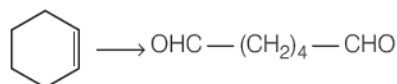
38 The nucleus of an atom can be assumed to be spherical. The radius of the nucleus of mass number A is given by  $1.25 \times 10^{-13} \times A^{1/3}$  cm. Radius of atom is 1 Å. If the mass number is 64, the fraction of the atomic volume that is occupied by the nucleus is

- (a)  $1.0 \times 10^{-3}$  (b)  $5.0 \times 10^{-3}$   
(c)  $2.5 \times 10^{-2}$  (d)  $1.25 \times 10^{-13}$

39 Carboxylic acids readily dissolve in aqueous sodium bicarbonate, liberating carbon dioxide. Which one of the following is correct statement?

- (a) Free carboxylic acid and its conjugate base are of comparable stability  
(b) The free carboxylic acid is more stable than its conjugate base  
(c) The conjugate base of the carboxylic acid is more stable than the free carboxylic acid  
(d) The conjugate acid of the carboxylic acid is more stable than the free carboxylic acid

40 Select the reagent for the following reaction,



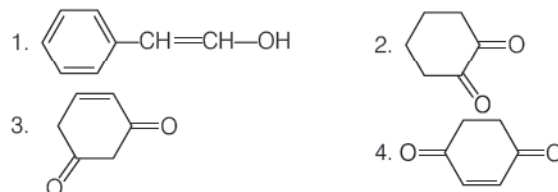
- (a)  $SeO_2$  (b)  $O_3, Zn/H_2O$   
(c)  $O_3, H_2O_2-CH_3COOH$  (d) PCC

**Direction** (Q. Nos. 41-42) In the following questions more than one of the answers given may be correct. Select the correct answers and mark it according to the codes.

**Codes**

- (a) 1, 2 and 3 are correct (b) 1 and 2 are correct  
(c) 2 and 4 are correct (d) 1 and 3 are correct

41 Tautomerism is exhibited by



42 Transition elements form binary compounds with halogens, which of the following elements will form  $MF_3$  type compounds? **→ [NCERT Exemplar]**

1. Cr 2. Co 3. Cu 4. Ni

43 Match the reaction (in Column I) with relation between  $\Delta H$  and  $\Delta E$  (in Column II) and choose the correct codes.

| Column I                                         | Column II                      |
|--------------------------------------------------|--------------------------------|
| A. $C(s) + O_2(g) \longrightarrow CO_2(g)$       | 1. $\Delta H = \Delta E + RT$  |
| B. $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$   | 2. $\Delta H = \Delta E - RT$  |
| C. $PCl_5(g) \longrightarrow PCl_3(g) + Cl_2(g)$ | 3. $\Delta H = \Delta E$       |
| D. $2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$  | 4. $\Delta H = \Delta E - 2RT$ |

**Codes**

- A B C D A B C D  
(a) 3 4 1 2 (b) 2 3 1 4  
(c) 3 1 4 2 (d) 4 2 3 1

**Direction** (Q. Nos. 44-45) Each of these questions contains two statements : Assertion and Reason. Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select one of the codes (a), (b), (c) and (d) given below.

- (a) Assertion is true, Reason is true; Reason is the correct explanation for Assertion  
(b) Assertion is true, Reason is true; Reason is not the correct explanation for Assertion  
(c) Assertion is true, Reason is false  
(d) Assertion is false, Reason is true

44 **Assertion**  $Zn^{2+}$  is diamagnetic.

**Reason** The electrons are lost from 4s orbital to form  $Zn^{2+}$ .

45 **Assertion** Cyclobutane is less stable than cyclopentane.

**Reason** Presence of bent bonds causes loss of orbital overlap.

## ANSWERS

|        |        |        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 1 (a)  | 2 (a)  | 3 (a)  | 4 (c)  | 5 (d)  | 6 (b)  | 7 (a)  | 8 (a)  | 9 (a)  | 10 (d) |
| 11 (d) | 12 (c) | 13 (b) | 14 (a) | 15 (b) | 16 (b) | 17 (a) | 18 (c) | 19 (b) | 20 (a) |
| 21 (a) | 22 (d) | 23 (d) | 24 (b) | 25 (b) | 26 (c) | 27 (c) | 28 (d) | 29 (d) | 30 (c) |
| 31 (c) | 32 (a) | 33 (b) | 34 (c) | 35 (d) | 36 (d) | 37 (d) | 38 (d) | 39 (c) | 40 (b) |
| 41 (a) | 42 (b) | 43 (a) | 44 (b) | 45 (c) |        |        |        |        |        |

## Hints and Explanations

**1** Larger the size of the halogen atom, lesser will be the back donation of electrons into empty  $2p$  orbital of B, i.e. more deficient is the molecule.

**2** Acidic nature of oxide is related to non-metallic nature of element. Non-metallic nature decreases in the order  $\text{Cl} > \text{S} > \text{P}$ .

**3**  $\text{Fe}_2\text{Cl}_6$  is a covalent compound in which Fe has only 6 electrons and hence, completes its octet by forming a coordinate bond with the chlorine atom of other molecule. Hence,  $\text{FeCl}_3$  exists as a dimer.

$$4 \left( p + \frac{a}{V^2} \right) (V - b) = RT$$

At high pressure,  $b$  cannot be neglected in comparison to  $V$ . Further, though  $V$  becomes small,  $a/V^2$  is large but as  $p$  is very high,  $\frac{a}{V^2}$  can be neglected in

$$p(V - b) = RT$$

$$\text{or } pV = RT + pb$$

(Dividing with  $RT$  in both sides)

$$\text{or } \frac{pV}{RT} = 1 + \frac{pb}{RT}$$

$$\text{or } Z = 1 + \frac{pb}{RT}$$

**5**  $\text{PCl}_3$  and  $\text{AsCl}_3$  have  $sp^3$  hybridisation and  $\text{PF}_5$  has  $sp^3d$  hybridisation. Hence, in group of  $\text{PCl}_3$ ,  $\text{AsCl}_3$  and  $\text{PF}_5$  all do not have  $sp^3d$  hybridisation.

**6** Nascent hydrogen is associated with chemical energy liberated in the reaction and hence, the molecules consist of excess energy.

**7** In metal carbonyls, the oxidation number of metal is always zero.

(Charge on Fe) + 5 (charge on CO) = 0

(Charge on Fe) + 5 (0) = 0

Charge on Fe = 0

(a) Oxidation state of Fe in  $\text{Fe}(\text{CO})_5$  is zero.

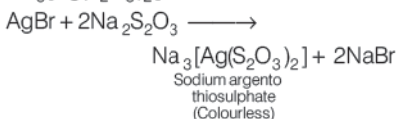
Similarly,

(b) Oxidation state of Fe in  $\text{Fe}_2\text{O}_3$  is +3.

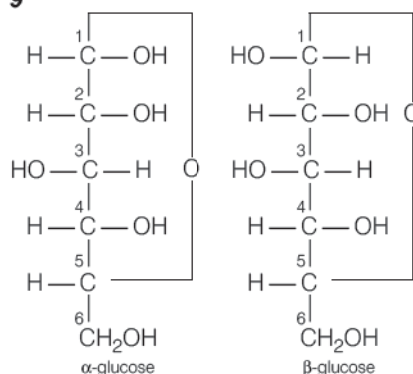
(c) Oxidation state of Fe in  $\text{K}_4[\text{Fe}(\text{CN})_6]$  is +2.

(d) Oxidation state of Fe in  $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$  is +2.

**8** The solubility of silver bromide in hypo solution is due to the formation of  $\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$ .



**9**

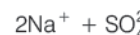
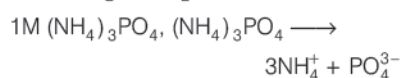
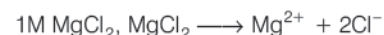


The two cyclic hemiacetal forms of glucose differ only in the configuration of the hydroxyl group at C1, called anomeric carbon (the aldehyde carbon before cyclisation). Such isomers, i.e.  $\alpha$ -form and  $\beta$ -forms are called anomers.

**10** As aluminium has more affinity for oxygen, it is used as a reducing agent for the extraction of Cr, Fe, Mn, etc.

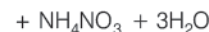
**11** Zr and Hf possess similar radii (Zr 160 pm, Hf 159 pm) and hence are called twins of periodic table. It is due to lanthanide contraction.

**12**  $1 \text{ M NaCl}, \text{NaCl} \longrightarrow \text{Na}^+ + \text{Cl}^-$



$(\text{NH}_4)_3\text{PO}_4$  dissociates to give maximum number of particles. Hence, its osmotic pressure will be maximum.

**13** The balanced redox equation is  $4\text{Fe} + 10\text{HNO}_3 \longrightarrow 4\text{Fe}(\text{NO}_3)_2$



Thus, the correct ratio of coefficients of  $\text{HNO}_3$ ,  $\text{Fe}(\text{NO}_3)_2$  and  $\text{NH}_4\text{NO}_3$  respectively is 10 : 4 : 1.

**14** Sublimation is : Solid  $\rightarrow$  Vapour. It can be written in two steps:

Solid  $\rightarrow$  Liquid  $\rightarrow$  Vapour

Solid  $\rightarrow$  Liquid requires enthalpy of fusion

Liquid  $\rightarrow$  Vapour requires enthalpy of vaporisation

**15** In case of optically active alkyl halides, the product formed as a result of  $\text{S}_\text{N}2$  mechanism which has the inverted configuration as compared to the reactant. This is because the nucleophile attached itself on the side opposite to the one where the halogen atom is present. Thus,  $\text{S}_\text{N}2$  reaction of optically active halides are accompanied by inversion of configuration.

If one of the enantiomer is dextro-rotatory the other will be laevorotatory. In case of dextro-rotatory optically active alkyl hydrolysis by  $S_N2$  mechanism the resulting alcohol will be laevorotatory.

$$16 \quad w = \frac{ENV}{1000} \text{ [For monobasic acid, } E = M]$$

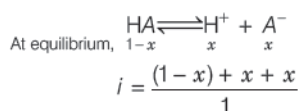
$$0.25 = \frac{M \times 1 \times 15.1}{1000 \times 10}$$

$$M = 165.56$$

$$\text{Molality } (m) = \frac{1 \times 1000}{165.56 \times 100} = 0.060$$

$$\Delta T_f = K_f \times m = 1.86 \times 0.060 = 0.1116$$

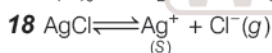
$$i = \frac{(\Delta T_f)_{\text{observed}}}{(\Delta T_f)_{\text{normal}}} = \frac{0.186}{0.1116} = 1.66$$



$$1.66 = 1 + x$$

$$x = 0.66$$

- 17 Both  $P_4O_{10}$  and  $P_2O_5$  may be regarded as anhydrides of  $H_3PO_4$  but since,  $P_2O_5$  occurs as a dimer, the right answer is  $P_4O_{10}$  and not  $P_2O_5$ .



Let solubility is  $S$  mol  $L^{-1}$  in 1 M KCl.

$$[Ag^+] = S$$

$$\therefore [Cl^-] = S + 1 \approx 1 \text{ M}$$

[ $\because$  For KCl,  $[Cl^-] = 1$ ]

$$[Ag^+][Cl^-] = K_{sp}$$

$$(S \times 1) = K_{sp} = 1 \times 10^{-10}$$

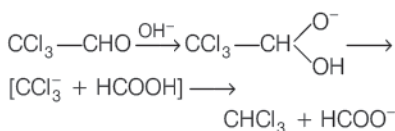
$$\text{or } S = 1 \times 10^{-10} \text{ M}$$

- 19 (a) In  $MnO_2$  and  $FeCl_3$ , oxidation states of Mn and Fe are +4 and +3, respectively.  
 (b) In  $[MnO_4]^-$ ,  $CrO_2Cl_2$ , oxidation states of Mn and Cr are +7 and +6, respectively.  
 (c) In  $[Fe(CN)_6]^{3-}$ ,  $[Co(CN)_3]$ , oxidation states of Fe and Co are +3 and +3, respectively.  
 (d)  $[NiCl_4]^{2-}$ ,  $[CoCl_4]^-$ , oxidation states of Ni and Co are +2 and +3, respectively.

Therefore,  $[MnO_4]^-$ ,  $CrO_2Cl_2$  have highest oxidation state pair of compounds

- 20 Although,  $CCl_3CHO$  does not contain  $\alpha$ -hydrogen yet it does not undergo

Cannizzaro reaction, since attack of  $OH^-$  ion cleaves the molecule to give  $CHCl_3$  and  $HCOO^-$ .



- 21 Amongst aldehydes and the acid derivatives, acid chlorides are the most susceptible to nucleophilic attack due to strong  $-I$ -effect and weak  $+R$ -effect of the Cl-atom, as a result of which carbonyl carbon has the highest electron deficiency. The actual order is  $MeCOCl > MeCOOCOMe > MeCOOMe > MeCHO$

- 22 As  $\Delta G^\circ > 0$ , the equilibrium constant  $K$  would be less than 1 ( $\Delta G^\circ = -RT \ln K$ ).

$$23 \quad 5.85 \text{ g NaCl} = \frac{5.85}{58.5} \text{ mol} = 0.1 \text{ mol}$$

$$90 \text{ g } H_2O = \frac{90}{18} \text{ mol} = 5 \text{ mol}$$

$\therefore$  Mole fraction of NaCl

$$= \frac{0.1}{5 + 0.1} = \frac{0.1}{5.1} = 0.0196$$

$$24 \quad \text{Bond order} = \frac{1}{2}(N_b - N_a)$$

$$\text{Bond order in } O_2 = \frac{1}{2}(10 - 6) = 2$$

$$\text{Bond order in } O_2^- = \frac{1}{2}(10 - 7) = 1.5$$

$$\text{Bond order in } O_2^+ = \frac{1}{2}(10 - 5) = 2.5$$

$$\text{Bond order in } O_2^{2-} = \frac{1}{2}(10 - 8) = 1$$

As the bond order increases, the bond strength increases accordingly. Therefore, the correct order is



- 25 When an ideal gas undergoes expansion, no work is done in overcoming attractive forces. Hence, internal energy does not fall and hence, temperature also does not fall.

$$26 \quad p_1 = p, V_1 = V, p_2 = 2p, V_2 = 2V$$

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}, \frac{pV}{T_1} = \frac{2p \times 2V}{T_2}, T_2 = 4T_1$$

When air has been taken in and  $p, V$  remains constant.

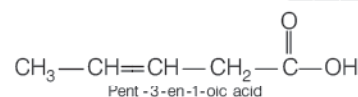
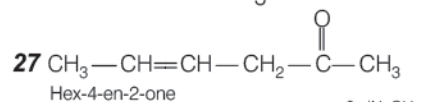
$$n_1 \cdot 4T_1 = n_2 \cdot T_2$$

$$n_1 = n$$

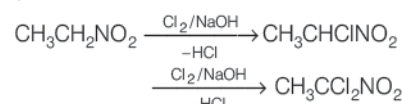
$$n_2 = n + \frac{1}{4}n = \frac{5}{4}n$$

$$n \cdot 4T_1 = \frac{5}{4}n \cdot T_2$$

$$T_2 = \frac{16}{5}T_1$$



- 28  $\alpha$ -H atoms in nitroalkanes being acidic are successively replaced by Cl in the presence of NaOH.

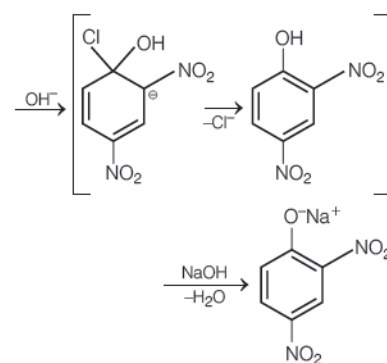
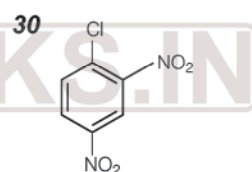


- 29 Arrhenius equation gives relation of rate constant with temperature.

$$k = A \cdot e^{-E_a/RT}$$

On taking logarithm, we get

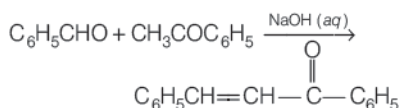
$$\ln k = \ln A - \frac{E_a}{RT} \ln e \text{ or } \ln k = \ln A - \frac{E_a}{RT}$$



This reaction occurs by activated nucleophilic substitution.

- 31 If  $CHCl_3$  sample contains phosgene ( $COCl_2$ ), it will give a white precipitate when treated with  $AgNO_3$  solution in cold. However,  $AgNO_3$  solution after boiling with alcoholic KOH will always give white precipitate with  $CHCl_3$ , whether it contains  $COCl_2$  or not.

- 32 In the presence of sodium hydroxide, benzaldehyde reacts with acetophenone, to give phenyl cinnamate.

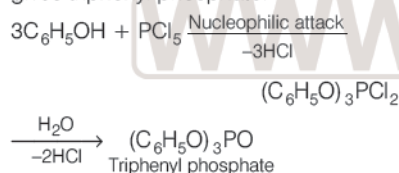


- 33** Since, the alkene on oxidation gives only acetic acid, therefore the alkene must be symmetrical containing two carbon atoms on either side of the double bond, i.e. 2-butene.

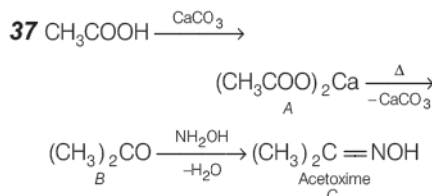
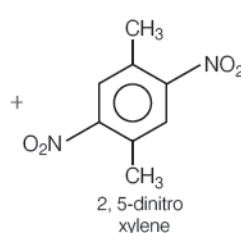
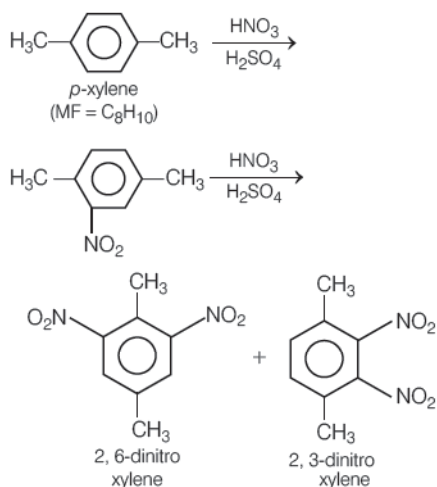


- 34** As the conjugation increases, heat of hydrogenation decreases. Thus, alkene (c) with two isolated double bonds has the highest heat of hydrogenation. Heat of hydrogenation is also an index of stability. The lower the heat of hydrogenation of an alkene the more stable it is.

- 35** Due to resonance, carbon-oxygen bond in phenol is quite stronger than in alcohols and hence, —OH group cannot be displaced by —Cl on treatment with  $\text{PCl}_5$ . Instead nucleophilic attack by —OH of phenol occurs on  $\text{PCl}_5$  with displacement of three Cl-atoms. The resulting product on hydrolysis gives triphenyl phosphate.



- 36** In *p*-xylene, the four nuclear H-atoms are equivalent and hence, only one mononitro derivative is formed. But it gives three dinitro derivatives (2, 3; 2, 6 and 2, 5) as shown below:



- 38** Radius of nucleus  
 $= 1.25 \times 10^{-13} \times A^{1/3} \text{ cm}$   
 $= 1.25 \times 10^{-13} \times (64)^{1/3}$   
 $= 1.25 \times 10^{-13} \times 4 \text{ cm}$   
 $= 5 \times 10^{-13} \text{ cm}$

$$\text{Radius of atom} = 1 \text{ \AA} = 10^{-8} \text{ cm}$$

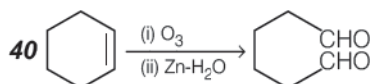
$$\frac{\text{Volume of nucleus}}{\text{Volume of atom}}$$

$$= \frac{(4/3)\pi(5 \times 10^{-13})^3}{(4/3)\pi(10^{-8})^3}$$

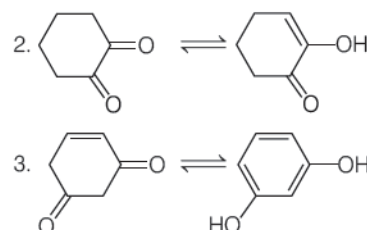
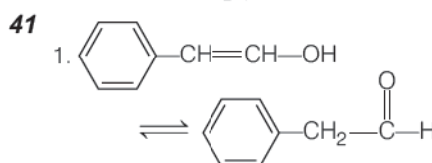
$$= \frac{125 \times 10^{-39}}{10^{-24}} = 125 \times 10^{-15}$$

$$= 1.25 \times 10^{-13}$$

- 39**  $\text{RCOOH} + \text{NaHCO}_3 \rightleftharpoons \text{RCOONa} + \text{H}_2\text{O} + \text{CO}_2$   
 or  $\text{RCOOH} + \text{HCO}_3^- \rightleftharpoons \text{RCOO}^- + \text{H}_2\text{O} + \text{CO}_2$   
 Conjugate base,  $\text{RCOO}^-$  is more stable due to resonance, than the free carboxylic acid. That's why, equilibrium shifts in the forward direction.

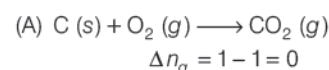


$\text{Zn-H}_2\text{O}$  is the reagent for reductive work up of ozonide.  $\text{H}_2\text{O}_2$ - $\text{CH}_3\text{COOH}$  would give  $\text{HOOC}-(\text{CH}_2)_4-\text{COOH}$ .

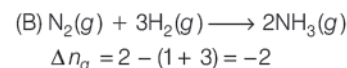


- 42** The +3 oxidation states is more stable only for Cr and Co. Cu exhibits only +1 and +2 oxidation states. Ni generally exhibits +2 oxidation state. This is the reason for the fact that Cr and Co form  $\text{MF}_3$  (where,  $M = \text{Cr}$  and  $\text{Co}$ ) type compounds whereas Cu and Ni do not.

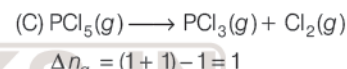
- 43**  $\Delta H = \Delta E + \Delta n_g RT$



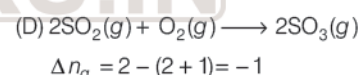
$$\therefore \Delta H = \Delta E$$



$$\therefore \Delta H = \Delta E - 2RT$$



$$\therefore \Delta H = \Delta E + RT$$



$$\therefore \Delta H = \Delta E - RT$$

- 44** Zn has outer electronic configuration  $3d^{10} 4s^2$ .

$\therefore \text{Zn}^{2+} = 3d^{10}$ , i.e. two electrons are removed from 4s orbital and it has zero unpaired electron. So, it is diamagnetic. The diamagnetic character is due to the pairing of all the electron spins, not due to the removal of electron.

- 45** For the stability of cycloalkanes, Baeyer's strain theory was given. According to this theory, if carbon atoms unit to form a closed chain, the valency angles are distorted their natural directions and a strain is set up in the molecules depending upon the extent of displacement of natural valency angles. The greater the strain, the greater is the instability of the ring. The value of distortion of valency angle in cyclobutane and cyclopentane is  $+9^\circ 34'$  and  $+0^\circ 54'$  respectively. Hence, cyclobutane is less stable.

## DAY FOURTY

# Mock Test 3

### Instructions

- The test consists of 45 questions.
- Candidates will be awarded marks for correct response of each question. 1/4 (one-fourth) marks will be deducted for indicating incorrect response of each question. No deduction from the total score will be made, if no response is indicated for an item in the answer sheet.
- There is only one correct response for each question. Filling up more than one response in each question will be treated as wrong response.

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- 1 3 moles of a perfect gas is expanded isothermally and reversibly from a volume of 1 L to a volume of 10 L at 27°C. The maximum work done is  
(a) 190 L atm (b) 170.17 L atm  
(c) 192.25 L atm (d) 185.23 L atm
- 2 Naturally occurring boron consists of two isotopes whose atomic weights are 10.01 and 11.01. The atomic weight of natural boron is 10.81. The percentage of each isotope in natural boron is  
(a) 20, 80 (b) 40, 60  
(c) 60, 40 (d) 80, 20
- 3 Arrange the following compounds in order of increasing dipole moment.  
I. Toluene II. *m*-dichlorobenzene  
III. *o*-dichlorobenzene IV. *p*-dichlorobenzene  
(a) I < IV < II < III (b) IV < I < II < III  
(c) IV < I < III < II (d) IV < II < I < III
- 4 If the molecule of HCl were totally polar, the expected value of dipole moment is 6.12 D (dbye), but the experimental value of dipole moment was 1.03 D. Calculate the percentage of ionic character.  
(a) 17 (b) 83 (c) 50 (d) Zero
- 5 AB is an ionic solid. The ionic radii of  $A^+$  and  $B^-$  are respectively  $r_c$  and  $r_a$ . Lattice energy of AB is proportional to  
(a)  $\frac{r_c}{r_a}$  (b)  $(r_c + r_a)$  (c)  $\frac{r_a}{r_c}$  (d)  $\frac{1}{(r_c + r_a)}$
- 6 1 mole of  $N_2H_4$  loses 10 moles of electrons to form a new compound y. Assuming that all nitrogen appears in the new compound, what is the oxidation state of nitrogen in y? (There is no change in the oxidation state of hydrogen)  
(a) +3 (b) -3 (c) -1 (d) +5
- 7 A compound formed by elements X and Y, crystallises in a cubic structure, where Y atoms are at the corner of the cube and X atoms are at alternate faces. The formula of the compound is  
(a)  $X_2Y$  (b) XY (c)  $XY_2$  (d)  $XY_3$
- 8 The relation between the three types of velocities is (where,  $\alpha$  = most probable velocity,  $v$  = average velocity and  $u$  = rms velocity)  
(a)  $u < v < \alpha$  (b)  $\alpha < v < u$   
(c)  $\alpha = v = u$  (d)  $\alpha \neq v = u$
- 9  $E^\circ_{Fe^{3+}/Fe} = -0.036$  V,  $E^\circ_{Fe^{2+}/Fe} = -0.439$  V. The value of standard electrode potential for the charge,  $Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$  will be  
(a) -0.072 V (b) 0.385 V  
(c) 0.770 V (d) -0.270 V
- 10 A solution of urea in water has a boiling point of 100.130°C. The freezing point of the same solution, ( $K_f$  and  $K_b$  for water are 1.87°C and 0.520°C, respectively) is  
(a) 0.520°C (b) 0.4675°C (c) -0.4675°C (d) -0.520°C

- 11 Which of the following reagents can be used to separate a mixture of aniline and phenol?

I. Water II. NaOH  
III. NaHCO<sub>3</sub> IV. HCl  
(a) I and II (b) II and III (c) I and IV (d) II and IV

- 12 The IUPAC name of the compound is

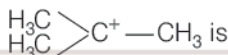


(a) bicyclo [2, 1, 0] pentane  
(b) 1,2-cyclopropyl cyclobutane  
(c) cyclopentane [4, 3] annulene  
(d) 1,2-methylene cyclobutane

- 13 Which is decreasing order of strength of bases?

$\text{OH}^-$ ,  $\text{NH}_2^-$ ,  $\text{HC} \equiv \text{C}^-$  and  $\text{CH}_3\text{CH}_2^-$   
(a)  $\text{H}_3\text{CCH}_2^- > \text{NH}_2^- > \text{HC} \equiv \text{C}^- > \text{OH}^-$   
(b)  $\text{HC} \equiv \text{C}^- > \text{CH}_3\text{CH}_2^- > \text{NH}_2^- > \text{OH}^-$   
(c)  $\text{OH}^- > \text{NH}_2^- > \text{CH} \equiv \text{C}^- > \text{H}_3\text{CCH}_2^-$   
(d)  $\text{NH}_2^- > \text{HC} \equiv \text{C}^- > \text{OH}^- > \text{H}_3\text{CCH}_2^-$

- 14 Among the following, the true property about

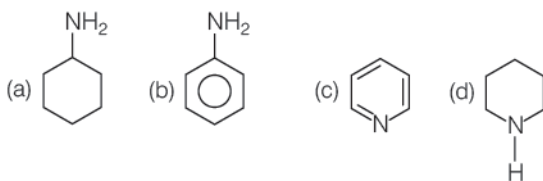


(a) it is non-planar  
(b) its C<sup>+</sup> is sp<sup>2</sup>-hybridised  
(c) an electrophile can attack on its C<sup>+</sup>  
(d) it does not undergo hydrolysis

- 15 Which of the following compounds exhibits stereoisomerism?

(a) 2-methylbutene-1 (b) 3-methylbutanoic acid  
(c) 3-methylbutyne-3 (d) 2-methylbutanoic acid

- 16 Which of the following is most basic?



- 17 The normality of orthophosphoric acid having purity of 70% by weight and specific gravity 1.54 is

(a) 11 N (b) 22 N  
(c) 33 N (d) 44 N

- 18 The composition of duralumin is

(a) Al-6%, Mg-94%  
(b) Cu-23%, Zn-56%, Mn-20%  
(c) Cu-50%, Al-40%, Si-10%  
(d) Al-95%, Cu-4%, Mn-0.5%, Mg-0.5%

- 19 Which of the following species is not expected to be a ligand?

[NCERT Exemplar]

(a) NO (b) NH<sub>4</sub><sup>+</sup>  
(c) NH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>NH<sub>2</sub> (d) CO

- 20 Total volume of atoms present in a face centred cubic unit cell of a metal is

(r = atomic radius)

(a)  $\frac{20}{3} \pi r^3$  (b)  $\frac{24}{3} \pi r^3$  (c)  $\frac{12}{3} \pi r^3$  (d)  $\frac{16}{3} \pi r^3$

- 21 Which is norbide?

(a) Silicon carbide (b) Boron carbide  
(c) Carbon carbide (d) Magnesium carbide

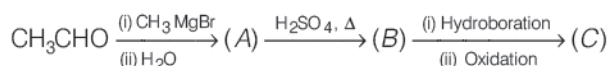
- 22 Epichlorohydrin is

(a) 3-chloropropane  
(b) 3-chloropropan-1-ol  
(c) 3-chloro-1,2-epoxypropane  
(d) None of the above

- 23 An organic compound of molecular formula C<sub>4</sub>H<sub>10</sub>O does not react with sodium. With excess of HI, it gives only one type of alkyl halide. The compound is

(a) ethoxyethane (b) 2-methoxypropane  
(c) 1-methoxypropane (d) 1-butanol

- 24 Compounds A and C in the following reaction are



[NCERT Exemplar]

(a) identical (b) positional isomers  
(c) functional isomers (d) optical isomers

- 25 Philosopher's wool when heated with BaO at 1100°C gives a compound. The compound is

(a) BaZnO<sub>2</sub> (b) Ba + ZnO<sub>2</sub>  
(c) BaCdO<sub>2</sub> (d) BaO<sub>2</sub> + Zn

- 26 Positive deviation from ideal behaviour takes place because of

(a) molecular interaction between atom and  $\frac{pV}{nRT} > 1$   
(b) molecular interaction between atom and  $\frac{pV}{nRT} < 1$   
(c) finite size of atoms and  $\frac{pV}{nRT} > 1$   
(d) finite size of atoms and  $\frac{pV}{nRT} < 1$

- 27 Which of the following is not a function of protein ?

(a) Nail formation  
(b) Skin formation  
(c) Muscle formation  
(d) Providing energy for metabolism

- 28 Which of the following does not represent the correct order of properties indicated?

(a) NH<sub>3</sub> < PH<sub>3</sub> < AsH<sub>3</sub> (acidic character)  
(b) Ni<sup>2+</sup> < Co<sup>2+</sup> < Fe<sup>2+</sup> < Mn<sup>2+</sup> (unpaired electrons)  
(c) Al<sub>2</sub>O<sub>3</sub> < MgO < Na<sub>2</sub>O < K<sub>2</sub>O (basic nature)  
(d) Li < Be < B < C (IE<sub>1</sub>)

- 29 K<sub>a</sub> for a weak monobasic acid is 1.0 × 10<sup>-6</sup>, the pK<sub>b</sub> of its conjugate base is

(a) 6.0 (b) 8.0 (c) 1 × 10<sup>-8</sup> (d) 1 × 10<sup>-6</sup>

**30** The pH of a buffer containing equal molar concentrations of a weak base and its chloride ( $K_b$  for weak base =  $2 \times 10^{-5}$ ,  $\log 2 = 0.3$ ) is  
(a) 5 (b) 9 (c) 4.7 (d) 9.3

**31** The time taken for 90% of a first order reaction to complete is approximately  
(a) 1.1 times that of half-life (b) 2.2 times that of half-life  
(c) 3.3 times that of half-life (d) 4.4 times that of half-life

**32** Which of the following requires catalyst?  
(a)  $S + O_2 \longrightarrow SO_2$  (b)  $2SO_2 + O_2 \longrightarrow 2SO_3$   
(c)  $C + O_2 \longrightarrow CO_2$  (d) All of these

**33** One mole of an non-ideal gas undergoes a change of state (2.0 atm, 3.0 L, 95 K  $\rightarrow$  4.0 atm, 5.0 L, 245 K) with a change in internal energy  $\Delta E = 30.0$  L atm. The enthalpy ( $\Delta H$ ) of the process is (in L atm)  
(a) 22 (b) 44 (c) 66 (d) 88

**34** For a reaction to be spontaneous at all temperatures  
(a)  $\Delta G$  and  $\Delta H$  should be negative  
(b)  $\Delta G$  and  $\Delta H$  should be positive  
(c)  $\Delta G = \Delta S = 0$   
(d)  $\Delta H < \Delta G$

**35** The cell constant of a conductivity cell

[NCERT Exemplar]

- (a) changes with change of electrolyte  
(b) changes with change of concentration of electrolyte  
(c) changes with temperature of electrolyte  
(d) remains constant for a cell

**36** The resistance of a 0.1 N solution of acetic acid is 250  $\Omega$  when measured in a cell of cell constant  $1.15 \text{ cm}^{-1}$ . The equivalent conductance (in  $\text{ohm}^{-1} \text{ cm}^2 \text{ equiv}^{-1}$ ) of 0.1 N acetic acid is  
(a) 46 (b) 9.2 (c) 18.4 (d) 2.3

**37** An alcohol gave Lucas test in about 5 min. When the alcohol was treated with hot concentrated  $H_2SO_4$ , it gave an alkene of molecular formula  $C_4H_8$ , which on ozonolysis gave  $C_2H_4O$ . The structure of alcohol is  
(a)  $CH_3CHOHCH_2CH_3$  (b)  $CH_3CH_2CH_2CH_2OH$   
(c)  $CH_3CHOHCH_2CH_2CH_3$  (d)  $(CH_3)_3C-OH$

**38** The lowest vapour pressure at  $25^\circ\text{C}$  is of  
(a)  $H_2O$  (b)  $CCl_4$  (c)  $CHCl_3$  (d)  $C_6H_6$

**39** The EAN of Zn in  $[Zn(OH)_4]^{2-}$  complex is  
(a) 16 (b) 26 (c) 36 (d) 46

**40** The ionisation energy of  $d$ -block elements of a given period vary only slightly from one another. This is because of  
(a) increase in nuclear charge  
(b) decrease in nuclear charge  
(c) increase in screening effect  
(d) Both (a) and (c)

**Direction** (Q. No. 41-42) In the following questions more than one of the answers given may be correct. Select the correct answers and mark it according to the codes.

**Codes**

- (a) 1, 2 and 3 are correct  
(b) 1 and 2 are correct  
(c) 2 and 4 are correct  
(d) 1 and 3 are correct

**41** 11.2 g of mixture of  $MCl$  (volatile) and  $NaCl$  gave 28.7 g of white precipitate with excess of  $AgNO_3$  solution. 11.2 g of same mixture on heating gave a gas that on passing into  $AgNO_3$  solution gave 14.35 g of white precipitate. Hence,

1. ionic mass of  $M^+$  is 18
2.  $MCl$  and  $NaCl$  are in the 1 : 2 molar ratio
3. mixture has equal mole fraction of  $MCl$  and  $NaCl$
4. ionic mass of  $M^+$  is 10

**42** A new carbon bond formation is possible in

1. Cannizzaro's reaction
2. Friedel-Crafts reaction
3. Clemmensen reaction
4. Reimer-Tiemann reaction

**43** Match the solid (in Column I) with its example (in Column II) and choose the correct codes given below.

| Column I           | Column II       |
|--------------------|-----------------|
| A. Ionic solid     | 1. C (graphite) |
| B. Metallic solid  | 2. Dry ice      |
| C. Covalent solid  | 3. NaCl         |
| D. Molecular solid | 4. Fe           |

**Codes**

- |     |   |   |   |   |     |   |   |   |   |
|-----|---|---|---|---|-----|---|---|---|---|
|     | A | B | C | D |     | A | B | C | D |
| (a) | 2 | 1 | 4 | 3 | (b) | 1 | 2 | 3 | 4 |
| (c) | 3 | 4 | 1 | 2 | (d) | 1 | 3 | 2 | 4 |

**Direction** (Q. No. 44-45) Each of these questions contains two statements : Assertion and Reason. Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select one of the codes (a), (b), (c) and (d) given below.

- (a) Assertion is true, Reason is true; Reason is the correct explanation for Assertion  
(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion  
(c) Assertion is true, Reason is false  
(d) Assertion is false, Reason is true

**44 Assertion** Liquid  $NH_3$  is used for refrigeration.  
**Reason** Liquid  $NH_3$  quickly vaporises.

**45 Assertion** Synthetic fibres like nylon-6, 6 are weak fibres.

**Reason** They have high molecular weight and high molecular size.

## ANSWERS

|        |        |        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 1 (b)  | 2 (a)  | 3 (b)  | 4 (a)  | 5 (d)  | 6 (a)  | 7 (b)  | 8 (b)  | 9 (c)  | 10 (c) |
| 11 (d) | 12 (a) | 13 (a) | 14 (b) | 15 (d) | 16 (d) | 17 (c) | 18 (d) | 19 (b) | 20 (d) |
| 21 (b) | 22 (c) | 23 (a) | 24 (b) | 25 (a) | 26 (a) | 27 (d) | 28 (d) | 29 (b) | 30 (d) |
| 31 (c) | 32 (b) | 33 (b) | 34 (a) | 35 (d) | 36 (a) | 37 (a) | 38 (a) | 39 (c) | 40 (d) |
| 41 (d) | 42 (c) | 43 (c) | 44 (a) | 45 (d) |        |        |        |        |        |

## Hints and Explanations

1  $W = 2.303 nRT \log_{10} \frac{V_2}{V_1}$   
 $= 2.303 \times 3 \times 0.0821 \times 300 \log_{10} \frac{10}{1}$   
 $= 170.17 \text{ L atm}$

2 Let % of isotope with atomic weight 10.01 =  $x$   
 % of isotope with atomic weight 11.01 =  $(100 - x)$

Since, atomic weight  
 $= \frac{x \times 10.01 + (100 - x) \times 11.01}{100}$   
 $10.81 = \frac{x \times 10.01 + (100 - x) \times 11.01}{100}$

$x = 20$

Hence, % of isotope with atomic weight 10.01 = 20

$\therefore$  % of isotope with atomic weight 11.01 =  $100 - 20 = 80$

3 Dipole moment of *p*-dichlorobenzene (IV) is zero due to symmetrical structure.  $\mu$  of C—CH<sub>3</sub> is the less than that of C—Cl. The vector sum of the two C—Cl bonds in the *o*-isomer (bond angle, 60°) is greater than that of the *m*-isomer (bond angle 120°).

4 Percentage ionic character  
 $= \frac{\text{experimental value of DM}}{\text{theoretical value of DM}} \times 100$   
 $= \frac{1.03}{6.12} \times 100 = 17\%$

5 The lattice becomes stronger (i.e. the lattice energy  $U$  becomes more negative), as  $r$  the interionic distance decreases.  $U$  is proportional to  $\frac{1}{r}$

or  $U \propto \frac{1}{(r_c + r_a)}$

6 Oxidation number of N in N<sub>2</sub>H<sub>4</sub> = -2

Increase in oxidation number of each N atom

$= \frac{\text{loss of moles of electrons}}{2}$

$= \frac{10}{2} = 5$

Hence, new oxidation number

$= -2 + 5 = +3$

7 Number of atoms Y at eight corners of a cubic unit cell

$= 8 \times \frac{1}{8} = 1$

Number of atoms X present on alternate face of unit cell

$= 2 \times \frac{1}{2} = 1$

$\therefore$  Formula of compound = XY

8 Relation between the three types of velocities :

$\alpha : v : u = \sqrt{\frac{2RT}{M}} : \sqrt{\frac{8RT}{\pi M}} : \sqrt{\frac{3RT}{M}}$   
 $= \sqrt{2} : \sqrt{\frac{8}{\pi}} : \sqrt{3}$   
 $= 1.414 : 1.595 : 1.734$   
 $= 1 : 1.128 : 1.234$

$\therefore \alpha < v < u$

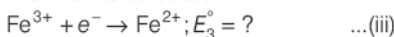
9 Given,  $\text{Fe}^{3+} + 3e^- \rightarrow \text{Fe}$ ;

$E_1^\circ = -0.036 \text{ V} \quad \dots\text{(i)}$



$E_2^\circ = -0.439 \text{ V} \quad \dots\text{(ii)}$

We need to calculate



We can obtain the (III) by subtracting II from I but  $E_3^\circ$ , we cannot obtain that way because electrode potential is intensive property. That's when we determine  $E_3^\circ$  calculating

$\Delta G_3 = \Delta G_1 - \Delta G_2 \quad (\Delta G \text{ is an extensive})$

$\Delta G_3 = 3 \times 0.036 \text{ F} - 2 \times 0.439 \text{ F}$

$\Delta G = 0.108 \text{ F} - 0.878 \text{ F}$

$-1 \times \text{F} \times E_3^\circ = -0.770 \text{ F}$

$E_3^\circ = 0.770 \text{ F}$

10  $\Delta T_b = 100.130 - 100 = 0.130^\circ\text{C}$

$K_b = 0.520^\circ\text{C}$

$\Delta T_b = \Delta K_b \times \text{molality of the solution}$

$\text{Molality} = \frac{\Delta T_b}{K_b} = \frac{0.130}{0.520} = 0.25$

$\Delta T_f = K_f \times \text{molality of the solution}$

$\Delta T_f = 1.87 \times 0.25 = 0.4675^\circ\text{C}$

Depression in freezing point,

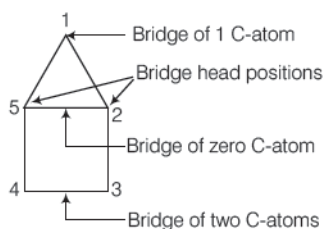
$\Delta T_f = \text{freezing point of solvent } (T_0) - \text{freezing point of solution } (T)$

$0.4675 = 0 - T$

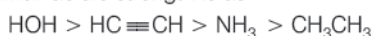
or,  $T = -0.4675^\circ\text{C}$

11 Aniline (a weak base) dissolves in HCl (a strong acid) while phenol (a weak base) dissolves in NaOH (a strong alkali). Hence, HCl and NaOH can be used for separating the mixture of aniline and phenol.

12 It is alicyclic compound with five carbon atoms, hence it is bicyclo pentane (prefix bicyclo is used because cycloalkane consists of two rings). The bridge head positions are tied together by three bridges of two, one and zero carbon atoms, hence its name is bicyclo (2, 1, 0) pentane.



- 13** The conjugate acids of  $\text{OH}^-$ ,  $\text{NH}_2^-$ ,  $\text{HC}\equiv\text{C}^-$  and  $\text{CH}_3\text{CH}_2^-$  are  $\text{H}_2\text{O}$ ,  $\text{NH}_3$ ,  $\text{HC}\equiv\text{CH}$  and  $\text{CH}_3-\text{CH}_3$  respectively. Their acidic strength is as

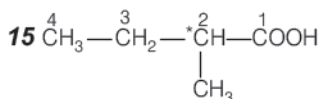


A strong acid has a weak conjugate base, hence the decreasing order of basic strength is



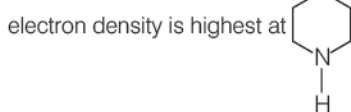
- 14** The carbon atom in carbocation has  $\text{sp}^2$ -hybridisation.

These carbocations have trigonal shape with positively charged carbon. Thus, the shape of  $\text{CH}_3^+$  may be considered as being derived from the overlap of three equivalent  $\text{C}(\text{sp}^2)$  hybridised orbitals with 1s orbital of each of the three hydrogen atoms. Each bond may be represented as  $\text{C}(\text{sp}^2)-\text{H}(1s)$  sigma bond.



Here,  $\text{C}_2$  is chiral. Hence, the molecule is optically active and will exhibit stereoisomerism.

- 16** Electron donors are bases. Since,



(piperidine), hence, it is most basic.

- 17** Normality

$$= \frac{\text{weight percentage} \times \text{density} \times 10}{\text{equivalent weight}}$$

$$= \frac{70 \times 1.54 \times 10}{98/3} = 33 \text{ N}$$

- 18** Duralumin, an alloy used for making aeroplanes, is composed of Al-95%, Cu-4%, Mn-0.5% and Mg-0.5%.

- 19** Ligand must donate a pair of electron or loosely held electron pair to metal and form a  $M-L$  bond.



$\text{NH}_4^+$  ion has no lone pair of electrons hence, it is not a ligand.

- 20** Volume of an atom =  $\frac{4}{3}\pi r^3$

In fcc, number of atoms per unit cell

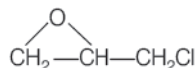
$$= 4$$

$$\therefore \text{Volume of total atoms} = 4 \times \frac{4}{3}\pi r^3$$

$$= \frac{16}{3}\pi r^3$$

- 21** Norbide is hot pressed boron carbide ( $\text{B}_4\text{C}$ ) which is one of the hardest materials known and offers excellent chemical and wear resistance, for demanding applications such as blast nozzles, monument nozzles and armor components.

- 22** Epichlorohydrin is

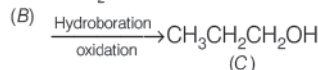
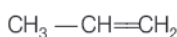
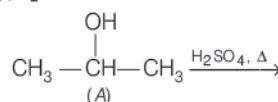


It is 3-chloro-1, 2-epoxypropane.

- 23** As a single alkyl halide is formed on treatment with HI, it must be  $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$  (ethoxyethane).

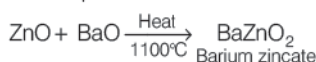


- 24**  $\text{CH}_3\text{CHO} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) CH}_3\text{MgBr}}$



(A), (B) and (C) compounds are positional isomers.

- 25** ZnO is commonly known as Philosopher's wool.



- 26**  $\frac{pV}{nRT} > 1$ , the gas is less compressible than expected from ideal behaviour and shows positive deviation.

- 27** Proteins do not provide energy for metabolism. Fibrous protein ionisation enthalpy keratin is responsible for the formation of nail, skin and myosin for muscle formation.

- 28** Generally, ionisation enthalpy increases in a period from left to right but ionisation enthalpy of II group is greater than III group due to stable electronic configuration of II group.

- 29**  $K_a$  and  $K_b$  are the ionisation constant of a weak acid and its conjugate base, then

$$K_a \cdot K_b = K_w = 1 \times 10^{-14}$$

$$K_b = \frac{1.0 \times 10^{-14}}{1.0 \times 10^{-6}} = 1.0 \times 10^{-8}$$

$$\text{p}K_b = -\log 1.0 \times 10^{-8} = 8$$

- 30**  $\text{pOH} = \text{p}K_b = -\log(2 \times 10^{-5})$

$$\left[ \because \log \frac{[\text{salt}]}{[\text{base}]} = 1 \right]$$

$$= 5 - 0.30 = 4.70$$

$$\text{pH} = 14 - 4.70 = 9.30$$

- 31**  $t_{90\%} = \frac{2.303}{k} \log \frac{a}{a - 0.9a}$

$$= \frac{2.303}{k} \log 10$$

$$\text{Again, } = \frac{2.303}{k}$$

$$t_{1/2} = \frac{2.303}{k} \log \frac{a}{a - 0.5a}$$

$$= \frac{2.303}{k} \log 2$$

$$= \frac{2.303}{k} \times 0.3010$$

$$\frac{t_{90\%}}{t_{1/2}} = \frac{1}{0.3010} = 3.3$$

or,  $t_{90\%} = 3.3$  times of  $t_{1/2}$

- 32** The reaction,  $2\text{SO}_2 + \text{O}_2 \longrightarrow 2\text{SO}_3$  has high energy of activation and requires a catalyst (either Pt or Mo) for lowering it down.

- 33**  $H = E + pV$ , since  $p$  is not constant.

$$\Delta H = (E_2 - E_1) + (p_2V_2 - p_1V_1)$$

$$= (30) + [(4 \times 5) - (2 \times 3)]$$

$$= 30 + [20 - 6] = 44 \text{ L atm}$$

- 34**  $\Delta G = \Delta H - T\Delta S$

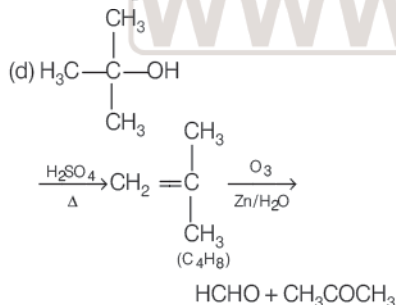
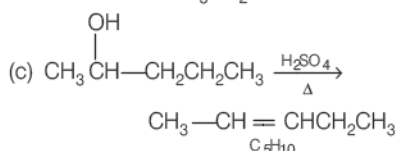
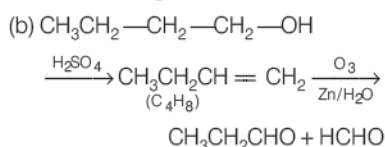
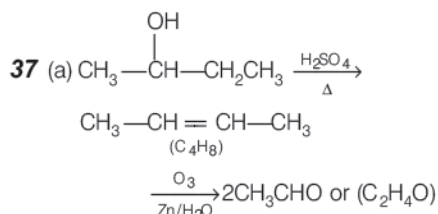
For a reaction to be spontaneous  $\Delta G$  must be negative. If  $\Delta G$  and  $\Delta H$  are negative, then reaction will be spontaneous at all temperature.

- 35** Cell constant remains constant for a cell. Cell constant is defined as the ratio of length of object and area of cross-section.  $a = \frac{l}{A}$ , since  $l$  and  $A$  remain constant.

$$36 \lambda_{eq} = \frac{1000k}{\text{normality}} = \frac{1000k}{0.1}$$

$$\left( \kappa = \frac{\text{cell constant}}{R} = \frac{1.15}{250} \right)$$

$$\text{So, } \lambda_{eq} = \frac{1000 \times 1.15}{250 \times 0.1} = 46$$



Secondary alcohol gives Lucas test in 5 min. Reaction sequence shows it is secondary alcohol.

38 Pure water (vapour pressure = 23.8 torr at 25°C)

39 The atomic number of Zn = 30  
Electrons contributed by Zn<sup>2+</sup> = 28

Electrons contributed by  
4OH<sup>-</sup> = 2 × 4 = 8

Thus, this complex follows Sidgwick's EAN rule. So, EAN = 28 + 8 = 36.

40 (a) Increase in nuclear charge and (c) increase in screening effect, are responsible for slightly higher I.E.  
∴ (d) is the correct option.

41 Let MCl = x g, NaCl = (11.2 - x) g

$$\text{MCl} + \text{AgNO}_3 \longrightarrow \text{MNO}_3 + \text{AgCl} \downarrow$$

White ppt

(M + 35.5) g MCl gives = 143.5 g AgCl

$$x \text{ g MCl gives } = \frac{143.5x}{M + 35.5} \text{ g AgCl ... (i)}$$

$$\text{NaCl} + \text{AgNO}_3 \longrightarrow \text{NaNO}_3 + \text{AgCl} \downarrow$$

(58.5 g) (143.5 g)

(11.2 - x) g NaCl gives

$$= \frac{143.5(11.2 - x)}{58.5} \text{ g AgCl ... (ii)}$$

$$\frac{143.5x}{M + 35.5} + \frac{143.5(11.2 - x)}{58.5} = 28.7$$

(given) ... (iii)

On heating MCl vaporises  
(being volatile)  
Thus, AgCl is formed due to MCl only

$$\text{Thus, } \frac{143.5x}{M + 35.5} = 14.35 \text{ ... (iv)}$$

Thus, so (iii),  $\frac{143.5(11.2 - x)}{58.5} = 14.35$

Thus, gives X = 5.35

From (IV), M = 18

MCl in mixture = 5.35 g

NaCl in mixture = 5.85 g

Thus, ionic mass of M<sup>+</sup> is 18.

Thus, 1 is correct.

Also, molar mass of MCl = 53.5 g mol<sup>-1</sup>

$$\text{moles of MCl} = \frac{5.35}{53.5} = 0.1$$

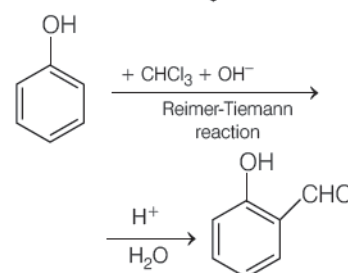
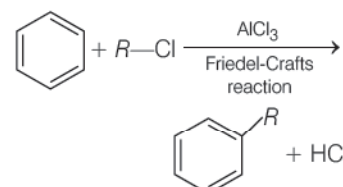
$$\text{moles of NaCl} = \frac{5.85}{58.5} = 0.1$$

$$\text{Mole fraction of MCl} = 0.5$$

Mole fraction of NaCl = 0.5

Thus, 3 is also correct.

42 In both Friedel-Crafts reaction and Reimer-Tiemann reaction, new carbon-carbon bond is formed.



| 43 S.No. | Column I        | Column II       |
|----------|-----------------|-----------------|
| A.       | Ionic solid     | 1. NaCl         |
| B.       | Metallic solid  | 2. Fe           |
| C.       | Covalent solid  | 3. C (graphite) |
| D.       | Molecular solid | 4. Dry ice      |

44 Liquid NH<sub>3</sub> is used for refrigeration because it vaporises quickly and for vaporisation, it takes up heat and cool the refrigerator.

45 Nylon 6, 6 is prepared by the condensation polymerisation of hexamethylene diamine with adipic acid under high pressure and at high temperature. Nylon 6,6 have high molecular weight and size, which is obtained by conducting the final stages of melt polymerisation. It is a strong fibre used in making sheets, bristles for brushes and in textile industry.

# NEET (National) SOLVED PAPER 2019

Held on : 5 May 2019

Duration : 3 Hours

Max. Marks : 180

## Important Instructions

- Test Booklet contains 45 questions.
- Each question carries 4 marks. For each correct response, the candidate will get 4 marks.
- For each incorrect response, one mark will be deducted from the total scores.
- The maximum marks are 180.

1. For the chemical reaction,  $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

The correct option is

- (a)  $-\frac{d[N_2]}{dt} = 2\frac{d[NH_3]}{dt}$  (b)  $-\frac{d[N_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$   
(c)  $3\frac{d[H_2]}{dt} = 2\frac{d[NH_3]}{dt}$  (d)  $-\frac{1}{3}\frac{d[H_2]}{dt} = -\frac{1}{2}\frac{d[NH_3]}{dt}$

2. The non-essential amino acid among the following is

- (a) leucine (b) alanine  
(c) lysine (d) valine

3. For an ideal solution, the correct option is

- (a)  $\Delta_{\text{mix}} V \neq 0$  at constant  $T$  and  $p$   
(b)  $\Delta_{\text{mix}} H = 0$  at constant  $T$  and  $p$   
(c)  $\Delta_{\text{mix}} G = 0$  at constant  $T$  and  $p$   
(d)  $\Delta_{\text{mix}} S = 0$  at constant  $T$  and  $p$

4. The manganate and permanganate ions are tetrahedral, due to

- (a) there is no  $\pi$ -bonding  
(b) the  $\pi$ -bonding involves overlap of  $p$ -orbitals of oxygen with  $p$ -orbitals of manganese  
(c) the  $\pi$ -bonding involves overlap of  $d$ -orbitals of oxygen with  $d$ -orbitals of manganese  
(d) the  $\pi$ -bonding involves overlap of  $p$ -orbitals of oxygen with  $d$ -orbitals of manganese

5. Which will make basic buffer?

- (a) 100 mL of 0.1 M  $CH_3COOH$  + 100 mL of 0.1 M  $NaOH$   
(b) 100 mL of 0.1 M  $HCl$  + 200 mL of 0.1 M  $NH_4OH$   
(c) 100 mL of 0.1 M  $HCl$  + 100 mL of 0.1 M  $NaOH$   
(d) 50 mL of 0.1 M  $NaOH$  + 25 mL of 0.1 M  $CH_3COOH$

6. The number of moles of hydrogen molecules required to produce 20 moles of ammonia through Haber's process is

- (a) 20 (b) 30 (c) 40 (d) 10

7. Which of the following diatomic molecular species has only  $\pi$ -bonds according to molecular orbital theory?

- (a)  $N_2$  (b)  $C_2$  (c)  $Be_2$  (d)  $O_2$

8. The biodegradable polymer is

- (a) nylon-2-nylon-6 (b) nylon-6  
(c) buna-S (d) nylon-6,6

9. Which of the following reactions are disproportionation reaction?

- (i)  $2Cu^+ \longrightarrow Cu^{2+} + Cu^0$   
(ii)  $3MnO_4^{2-} + 4H^+ \longrightarrow 2MnO_4^- + MnO_2 + 2H_2O$   
(iii)  $2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$   
(iv)  $2MnO_4^- + 3Mn^{2+} + 2H_2O \longrightarrow 5MnO_2 + 4H^+$

Select the correct option from the following.

- (a) (i), (ii) and (iii) (b) (i), (iii) and (iv)  
(c) (i) and (iv) only (d) (i) and (ii) only

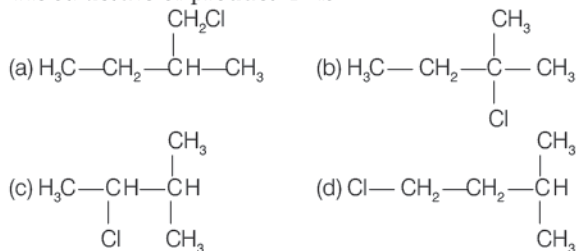
10. The compound that is most difficult to protonate is

- (a)  $H_3C-O-H$  (b)  $H_3C-O-CH_3$   
(c)  $Ph-O-H$  (d)  $H-O-H$

11. Which of the following species is not stable?

- (a)  $[GeCl_6]^{2-}$  (b)  $[Sn(OH)_6]^{2-}$  (c)  $[SiCl_6]^{2-}$  (d)  $[SiF_6]^{2-}$

12. An alkene 'A' on reaction with  $O_3$  and  $Zn - H_2O$  gives propanone and ethanal in equimolar ratio. Addition of HCl to alkene 'A' gives 'B' as the major product. The structure of product 'B' is



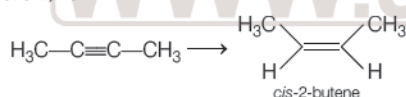
13. Match the following :

|                     |                                   |
|---------------------|-----------------------------------|
| A. Pure nitrogen    | (i) Chlorine                      |
| B. Haber process    | (ii) Sulphuric acid               |
| C. Contact process  | (iii) Ammonia                     |
| D. Deacon's process | (iv) Sodium azide or barium azide |

Which of the following is the correct option?

|          |       |      |       |           |      |       |      |
|----------|-------|------|-------|-----------|------|-------|------|
| A        | B     | C    | D     | A         | B    | C     | D    |
| (a) (ii) | (iv)  | (i)  | (iii) | (b) (iii) | (iv) | (ii)  | (i)  |
| (c) (iv) | (iii) | (ii) | (i)   | (d) (i)   | (ii) | (iii) | (iv) |

14. The most suitable reagent for the following conversion, is



- (a)  $H_2$ , Pd/C, quinoline (b)  $Zn/HCl$   
(c)  $Hg^{2+}/H^+$ ,  $H_2O$  (d)  $Na/liquid\ NH_3$

15. The correct order of the basic strength of methyl substituted amines in aqueous solution is

- (a)  $(CH_3)_3N > CH_3NH_2 > (CH_3)_2NH$   
(b)  $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2$   
(c)  $CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N$   
(d)  $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N$

16. A gas at 350 K and 15 bar has molar volume 20 percent smaller than that for an ideal gas under the same conditions. The correct option about the gas and its compressibility factor (Z) is

- (a)  $Z > 1$  and repulsive forces are dominant  
(b)  $Z < 1$  and attractive forces are dominant  
(c)  $Z < 1$  and repulsive forces are dominant  
(d)  $Z > 1$  and attractive forces are dominant

17. The number of sigma ( $\sigma$ ) and pi ( $\pi$ ) bonds in pent-2-en-4-yne is

- (a) 8 $\sigma$ -bonds and 5 $\pi$ -bonds (b) 11 $\sigma$ -bonds and 2 $\pi$ -bonds  
(c) 13 $\sigma$ -bonds and no  $\pi$ -bonds (d) 10 $\sigma$ -bonds and 3 $\pi$ -bonds

18. For the second period elements the correct increasing order of first ionisation enthalpy is

- (a)  $Li < B < Be < C < O < N < F < Ne$   
(b)  $Li < B < Be < C < N < O < F < Ne$   
(c)  $Li < Be < B < C < O < N < F < Ne$   
(d)  $Li < Be < B < C < N < O < F < Ne$

19. Among the following, the narrow spectrum antibiotic is

- (a) ampicillin (b) amoxycillin  
(c) chloramphenicol (d) penicillin G

20. The method used to remove temporary hardness of water is

- (a) Clark's method (b) ion-exchange method  
(c) synthetic resins method (d) Calgon's method

21. Which of the following series of transitions in the spectrum of hydrogen atom fall in visible region?

- (a) Balmer series (b) Paschen series  
(c) Brackett series (d) Lyman series

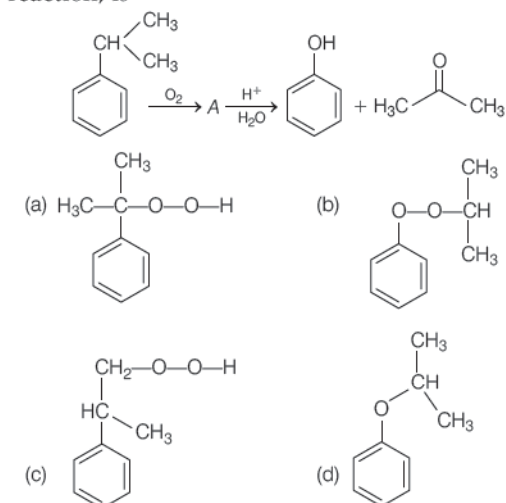
22. Match the xenon compounds in Column I with its structure in Column II and assign the correct code:

| Column I    | Column II                  |
|-------------|----------------------------|
| A. $XeF_4$  | (i) Pyramidal              |
| B. $XeF_6$  | (ii) Square planar         |
| C. $XeOF_4$ | (iii) Distorted octahedral |
| D. $XeO_3$  | (iv) Square pyramidal      |

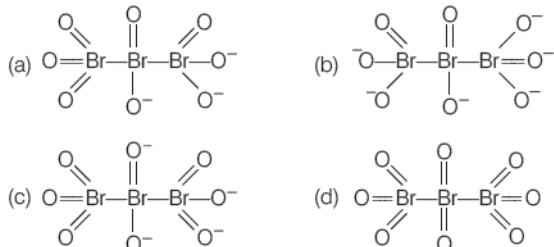
Codes

|           |       |      |      |          |       |       |      |
|-----------|-------|------|------|----------|-------|-------|------|
| A         | B     | C    | D    | A        | B     | C     | D    |
| (a) (ii)  | (iii) | (iv) | (i)  | (b) (ii) | (iii) | (i)   | (iv) |
| (c) (iii) | (iv)  | (i)  | (ii) | (d) (i)  | (ii)  | (iii) | (iv) |

23. The structure of intermediate A in the following reaction, is



24. The correct structure of tribromooctaoxide is



25. Which is the correct thermal stability order for  $H_2E$  ( $E = O, S, Se, Te$  and  $Po$ )?

- (a)  $H_2O < H_2S < H_2Se < H_2Te < H_2Po$   
 (b)  $H_2Po < H_2Te < H_2Se < H_2S < H_2O$   
 (c)  $H_2Se < H_2Te < H_2Po < H_2O < H_2S$   
 (d)  $H_2S < H_2O < H_2Se < H_2Te < H_2Po$

26. In which case change in entropy is negative?

- (a) Expansion of a gas at constant temperature  
 (b) Sublimation of solid to gas  
 (c)  $2H(g) \rightarrow H_2(g)$   
 (d) Evaporation of water

27. A compound is formed by cation  $C$  and anion  $A$ . The anions form hexagonal close packed (hcp) lattice and the cations occupy 75% of octahedral voids. The formula of the compound is

- (a)  $C_3A_2$  (b)  $C_3A_4$  (c)  $C_4A_3$  (d)  $C_2A_3$

28. Among the following, the one that is not a green house gas is

- (a) methane (b) ozone  
 (c) sulphur dioxide (d) nitrous oxide

29. Identify the incorrect statement related to  $PCl_5$  from the following:

- (a) Two axial P—Cl bonds make an angle of  $180^\circ$  with each other  
 (b) Axial P—Cl bonds are longer than equatorial P—Cl bonds  
 (c)  $PCl_5$  molecule is non-reactive  
 (d) Three equatorial P—Cl bonds make an angle of  $120^\circ$  with each other

30. Which of the following is an amphoteric hydroxide?

- (a)  $Ca(OH)_2$  (b)  $Mg(OH)_2$  (c)  $Be(OH)_2$  (d)  $Sr(OH)_2$

31.  $4d, 5p, 5f$  and  $6p$ -orbitals are arranged in the order of decreasing energy. The correct option is

- (a)  $6p > 5f > 5p > 4d$  (b)  $5p > 5f > 4d > 5p$   
 (c)  $5f > 6p > 4d > 5p$  (d)  $5f > 6p > 5p > 4d$

32. For the cell reaction,



$E^\circ_{\text{cell}} = 0.24 \text{ V}$  at 298 K. The standard Gibbs energy ( $\Delta_r G^\circ$ ) of the cell reaction is  
 [Given that Faraday constant  $F = 96500 \text{ C mol}^{-1}$ ]

- (a)  $-23.16 \text{ kJ mol}^{-1}$  (b)  $46.32 \text{ kJ mol}^{-1}$   
 (c)  $23.16 \text{ kJ mol}^{-1}$  (d)  $-46.32 \text{ kJ mol}^{-1}$

33. Conjugate base for Bronsted acids  $H_2O$  and  $HF$  are

- (a)  $H_3O^+$  and  $F^-$ , respectively (b)  $OH^-$  and  $F^-$ , respectively  
 (c)  $H_3O^+$  and  $H_2F^+$ , respectively (d)  $OH^-$  and  $H_2F^+$ , respectively

34. Which one is malachite from the following?

- (a)  $Cu(OH)_2$  (b)  $Fe_3O_4$   
 (c)  $CuCO_3 \cdot Cu(OH)_2$  (d)  $CuFeS_2$

35. pH of a saturated solution of  $Ca(OH)_2$  is 9.

The solubility product ( $K_{sp}$ ) of  $Ca(OH)_2$  is

- (a)  $0.25 \times 10^{-10}$  (b)  $0.125 \times 10^{-15}$   
 (c)  $0.5 \times 10^{-10}$  (d)  $0.5 \times 10^{-15}$

36. Which of the following is incorrect statement?

- (a)  $SiCl_4$  is easily hydrolysed  
 (b)  $GeX_4$  ( $X = F, Cl, Br, I$ ) is more stable than  $GeX_2$   
 (c)  $SnF_4$  is ionic in nature  
 (d)  $PbF_4$  is covalent in nature

37. Which mixture of the solutions will lead to the formation of negatively charged colloidal  $[AgI]I^-$  sol?

- (a) 50 mL of 1 M  $AgNO_3$  + 50 mL of 2 M KI  
 (b) 50 mL of 2 M  $AgNO_3$  + 50 mL of 1.5 M KI  
 (c) 50 mL of 1 M  $AgNO_3$  + 50 mL of 0.1 M KI  
 (d) 50 mL of 1 M  $AgNO_3$  + 50 mL of 1.5 M KI

38. The mixture that forms maximum boiling azeotrope is

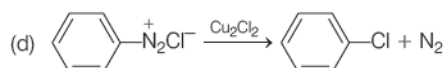
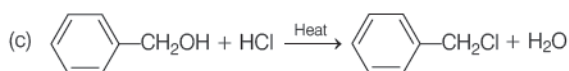
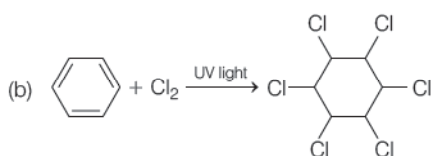
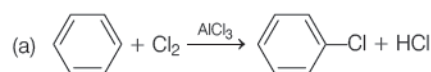
- (a) ethanol + water  
 (b) acetone + carbon disulphide  
 (c) heptane + octane  
 (d) water + nitric acid

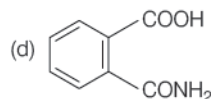
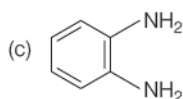
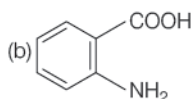
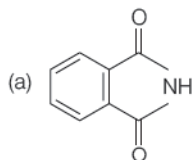
39. For a cell involving one electron,  $E^\circ_{\text{cell}} = 0.59 \text{ V}$  at 298 K, the equilibrium constant for the cell reaction is

[Given that,  $\frac{2.303 RT}{F} = 0.059 \text{ V}$  at  $T = 298 \text{ K}$ ]

- (a)  $1.0 \times 10^5$  (b)  $1.0 \times 10^{10}$   
 (c)  $1.0 \times 10^{30}$  (d)  $1.0 \times 10^2$

40. Among the following, the reaction that proceeds through an electrophilic substitution, is



$$\text{C}_6\text{H}_4(\text{COOH})_2 + \text{NH}_3 \xrightarrow{\text{Strong heating}}$$

$$(a) t_{2g}^6 e_g^0$$

(b)  $e^3 t_2^3$

(c)  $e^4 t_2^2$

(d)  $t_{2g}^4 e_g^2$

(a)  $t = 6.909/k$

(b)  $t = 4.606/k$

(c)  $t = 2.303/k$

(d)  $t = 0.693/k$

(a) 5 kJ

(b) 25 J

(c) 30 J

(d)  $-30 \text{ J}$

(a) Mg

(b) Ca

(c) Sr

(d) Be

## Answers

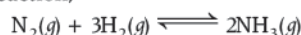
[illegible]

## Answer with Explanations

**1. (b)** Rate of reaction,

$$= \frac{\text{Decrease in the concentration of reactants or increase in the concentration of products}}{\text{Time taken}}$$

For the given reaction,



$$\begin{aligned} \text{Rate of reaction} &= \frac{-d[\text{N}_2]}{dt} \\ &= -\frac{1}{3} \frac{d[\text{H}_2]}{dt} = +\frac{1}{2} \frac{d[\text{NH}_3]}{dt} \end{aligned}$$

Hence, option (b) is correct.

**2. (b)** The amino acids that can be synthesised in our body and hence are not essentially required in our diet are called non-essential amino acids. e.g. glycine, alanine, serine, proline, cysteine, glutamine, tyrosine, aspartic acid, glutamic acid, asparagine. Rest given options are essential amino acids, i.e. those can't be synthesised in our body hence essentially required in our diet. Their other examples are isoleucine, phenylalanine, methionine, tryptophan, threonine, arginine and histidine.

Hence, option (b) is correct.

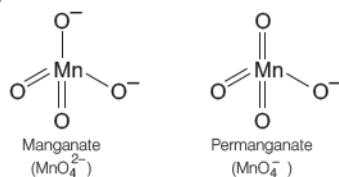
**3. (b)** Ideal solutions are those which obey Raoult's law over all concentration ranges at a given temperature, e.g. benzene-toluene, *n*-hexane-*n*-heptane, etc.

For an ideal solution,

$$\begin{aligned} \Delta V_{\text{mix}} &= 0, \Delta H_{\text{mix}} = 0, \\ \Delta G_{\text{mix}} &< 0, \Delta S_{\text{mix}} > 0. \end{aligned}$$

Hence, option (b) is correct.

**4. (d)** The structures of manganate and permanganate ions are as follows:



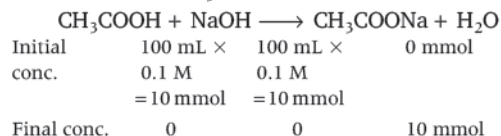
The manganate and permanganate ions are tetrahedral and contain  $\pi$ -bonds of  $d\pi$ - $p\pi$ -type, oxygen does not have any  $d$ -orbital.

Hence, the  $\pi$ -bonding takes place by overlap of  $p$ -orbital of oxygen and  $d$ -orbitals of manganese.

**5. (b)** **Key Idea** A buffer solution having pH more than 7 is known as basic buffer. It is obtained by mixing weak base and its salt with strong acid in a fixed proportion.

Let us consider all the options,

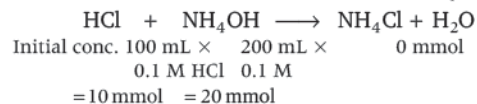
(a) 100 mL of 0.1 M  $\text{CH}_3\text{COOH}$  + 100 mL of 0.1 M  $\text{NaOH}$



It is not basic buffer because hydrolysis of salt takes place and final solution contains salt of weak acid with strong base only.

Hence, option (a) is incorrect.

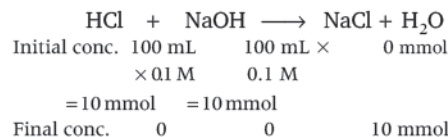
(b) 100 mL of 0.1 M  $\text{HCl}$  + 200 mL of 0.1 M  $\text{NH}_4\text{OH}$



Final conc. 0 10 mmol 10 mmol

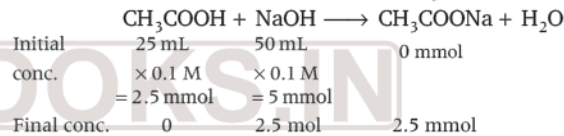
It is basic buffer because final solution contains weak base and its salt with strong acid. Hence, option (b) is correct.

(c) 100 mL of 0.1 M  $\text{HCl}$  + 100 mL of 0.1 M  $\text{NaOH}$



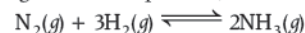
It is a neutral solution. Hence, option (c) is incorrect.

(d) 50 mL of 0.1 M  $\text{NaOH}$  + 25 mL of 0.1 M  $\text{CH}_3\text{COOH}$



It is basic solution. Hence, option (d) is incorrect.

**6. (b)** According to Haber's process,



Now, according to above equations

2 moles of ammonia ( $\text{NH}_3$ ) require = 3 moles of  $\text{H}_2$

$\therefore$  1 mole of  $\text{NH}_3$  require =  $3/2$  moles of  $\text{H}_2$

or, 20 moles of  $\text{NH}_3$  require =  $\frac{3}{2} \times 20$  moles of  $\text{H}_2$   
= 30 moles of  $\text{H}_2$ .

**Note** Involvement of any limiting reagent is not mentioned in question.

**7. (b)** The molecular orbital configuration of  $\text{C}_2$  is

$$\text{C}_2(Z=12) = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2 = \pi 2p_y^2.$$

Double bond in  $\text{C}_2$  consists of both  $\pi$ -bonds because of the presence of last (valence) four electrons in two  $\pi$ -molecular orbitals.

The configuration of  $\text{N}_2$ ,  $\text{Be}_2$  and  $\text{O}_2$  are as follows:

$$\text{N}_2(Z=14) = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2 = \pi 2p_y^2, \sigma 2p_z^2 \text{ (1 } \sigma \text{ and 2 } \pi\text{-bonds)}$$

$$\text{Be}_2(Z=8) = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2 \text{ (}\sigma \text{ bonds only)}$$

$$\text{O}_2(Z=16) = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma^* 2p_z^2, \pi 2p_x^2 = \pi 2p_y^2,$$

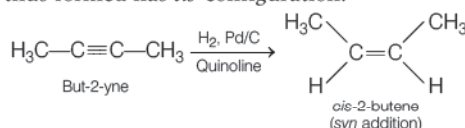
$$\pi^* 2p_x^1 = \pi^* 2p_y^1 \text{ (}\sigma, \pi \text{ and 1 } \pi^* \text{ bond)}$$

Hence, option (b) is correct.

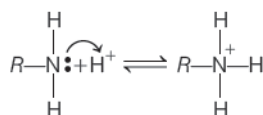
- (A)  $\rightarrow$  (iv), (B)  $\rightarrow$  (iii), (C)  $\rightarrow$  (ii), (D)  $\rightarrow$  (i)

**Time Saver** If someone know correct match of B, i.e. (iii) then according to the options (c) is the only correct answer as this match is given in this option only.

14. (a) Hydrogenation of alkynes in the presence of Pd/C, quinoline proceeds through *syn* addition of hydrogen and the alkene thus formed has *cis*-configuration.

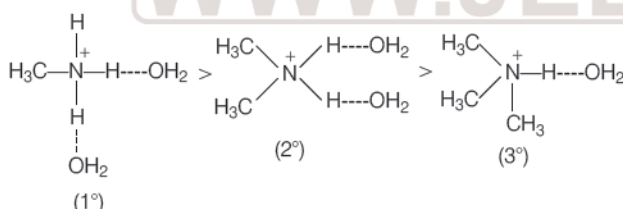


15. (d) Basic strength of methyl substituted amines in aqueous solution depends upon the ease of formation of cation by accepting a proton from the acid.



The basic strength can be decided by both inductive effect and solvation effect of alkyl group. In aqueous phase, the substituted  $\text{NH}_4^+$  cations get stabilised not only by electron releasing effect of alkyl group (+I) but also by solvation with water molecules. Greater the size of ion, lesser will be solvation and less stabilised is the ion. The order is as follows:

**Inductive effect**  $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)\text{NH}_2$



Strong +I-effect and hydrogen bonding favours higher basic strength of 2° amine. Thus, the correct order of basic strength in aqueous media will be  $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N}$

16. (b) Compressibility factor (Z) is the factor which decides the extent of deviation of real gases from ideal gases.

$$Z = \frac{V_{\text{real}}}{V_{\text{ideal}}}$$

Real gases deviate from ideal gas behaviour because for ideal gas it is considered that there is no force of attraction between gas molecules.

Also, for ideal gas, volume of gas molecules is negligible as compared to volume of gas container.

Given,  $V_{\text{real}} < V_{\text{ideal}}$

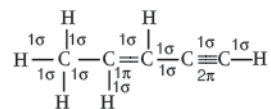
$\therefore Z < 1$

If  $Z < 1$ , the gas is more compressible than expected from ideal behaviour. As a result, attractive forces are present between molecules and are dominant.

17. (d) Single bond =  $1\sigma$ , Double bond =  $1\sigma + 1\pi$ ,

Triple bond =  $1\sigma + 2\pi$ .

The structure of pent-2-en-4-yne is



$\therefore$  The number of sigma ( $\sigma$ ) bonds are 10 and pi ( $\pi$ ) bonds are 3.

18. (a) The increasing nuclear charge outweighs the shielding effect across a period. As a result, the outermost electrons are held more firmly, hence ionisation enthalpy (IE) increases across a period as:

$$\text{Li} < \text{B} < \text{Be} < \text{C} < \text{O} < \text{N} < \text{F} < \text{Ne}$$

**Exceptions** (i) IE of boron (B) is less than IE of beryllium (Be). It is because presence of fully-filled 2s-orbitals in Be.

(ii) IE of oxygen (O) is less than that of nitrogen (N). It is because of the presence of half-filled p-orbitals in N which provides extra stability.

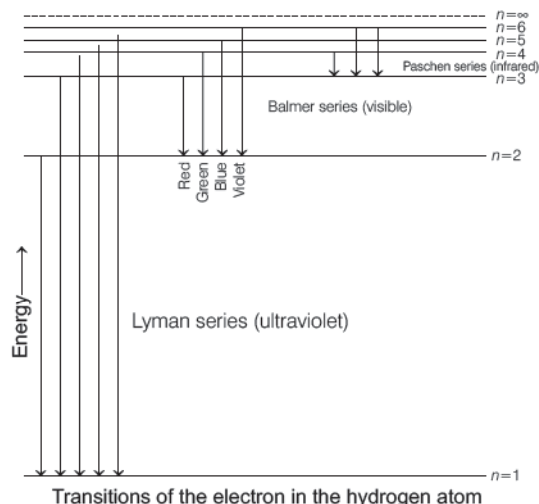
19. (d) **Key Idea** Antibiotics which are effective mainly against either gram positive or gram negative bacteria are called narrow spectrum antibiotics.

Penicillin G has a narrow spectrum. Ampicillin and amoxycillin are synthetic modifications of penicillin. These have broad spectrum. Also, chloramphenicol is a broad spectrum antibiotic.

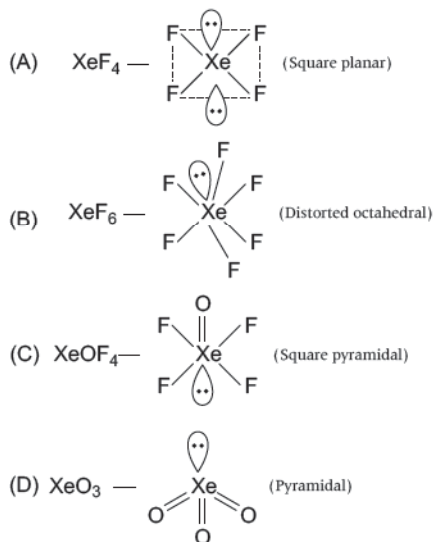
Hence, option (d) is correct.

20. (a) Temporary hardness in water is due to presence of magnesium and calcium hydrogen carbonates. Temporary hardness in water can be removed by Clark's method. In this method calculated amount of lime is added to hard water. It precipitates out calcium carbonate and magnesium hydroxide which can be filtered off.
- $$\text{Ca}(\text{HCO}_3)_2 + \text{Ca}(\text{OH})_2 \longrightarrow 2\text{CaCO}_3 \downarrow + 2\text{H}_2\text{O}$$
- $$\text{Mg}(\text{HCO}_3)_2 + 2\text{Ca}(\text{OH})_2 \longrightarrow 2\text{CaCO}_3 \downarrow + \text{Mg}(\text{OH})_2 \downarrow + 2\text{H}_2\text{O}$$
- Besides this, temporary hardness can also be removed by boiling. All the other given methods are used to remove permanent hardness of water.

21. (a) Balmer series of transitions in the spectrum of hydrogen atom fall in visible region. Lyman series fall in ultraviolet while paschen, brackett and pfund fall in infrared region.



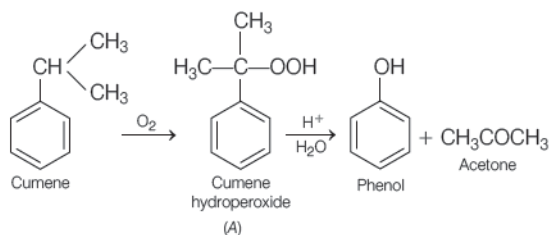
22. (a) The given xenon compounds with its structures are as follows:



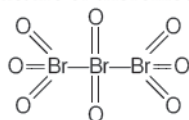
Hence, the correct match is

(A)  $\rightarrow$  (ii), (B)  $\rightarrow$  (iii), (C)  $\rightarrow$  (iv), (D)  $\rightarrow$  (i)

23. (a) The given reaction is of cumene process for phenol production and intermediate (A) is cumene hydroperoxide. In the process, cumene (isopropylbenzene) is oxidised in the presence of air to cumene hydroperoxide. Which is then converted to phenol and acetone by treating with dilute acid. Acetone, a by-product of this reaction is also obtained in large quantities by this method. The reaction takes place as follows:



24. (d) The correct structure of tribromooctaoxide is



In this compound, Br exhibits variable oxidation state. The oxidation state of Br from left to right are +6, +4 and +6, respectively.

25. (b) For group 16 elements, the hydrides with high molar mass (e.g.  $\text{H}_2\text{Po}$ ) are less thermally stable than hydride with lower molar mass (e.g.  $\text{H}_2\text{O}$ ). This is due to the increase in size of central atom which results to the weakening of  $M-H$  bond due to increased bond length.

Thus, the correct order of thermal stability for  $\text{H}_2\text{E}$  ( $\text{E} = \text{O}, \text{S}, \text{Se}, \text{Te}$  and  $\text{Po}$ ) is as follows:



26. (c) The explanation of given statements are :

- (a) Entropy is positive in case of expansion of a gas at constant temperature. It is because during expansion of gas volume increases and hence randomness increases.
- (b) Entropy is positive in case of sublimation of solid to gas as gas is much disordered than a solid.
- (c) Entropy is negative in case of  $2\text{H}(g) \rightarrow \text{H}_2(g)$  as the number of moles of gaseous reactants are more than that of gaseous products.
- (d) Entropy is positive in case of evaporation of water as gas is much disordered than a liquid.
- Hence, option (c) is correct.

27. (b) Anions (A) form hexagonal close packed (hcp) lattice, so

Number of anions (A) = 6

Number of octahedral voids = Number of atoms in the close packed structure = 6.

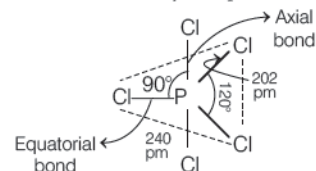
Cations (C) occupy 75% of octahedral voids, so number of cations (C) =  $6 \times \frac{75}{100} = 6 \times \frac{3}{4} = 9/2$

$\therefore$  The formula of compound =  $\text{C}_{9/2}\text{A}_6 = \text{C}_9\text{A}_{12} = \text{C}_3\text{A}_4$

Thus, option (b) is correct.

28. (c) Sulphur dioxide ( $\text{SO}_2$ ) is not a green house gas. Carbon dioxide ( $\text{CO}_2$ ), methane ( $\text{CH}_4$ ), water vapour, nitrous oxide ( $\text{N}_2\text{O}$ ), CFCs and ozone ( $\text{O}_3$ ) are the green house gases. These gases are responsible for global warming.

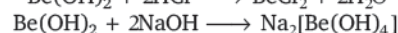
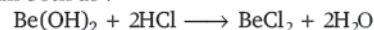
29. (c) In gaseous and liquid phases,  $\text{PCl}_5$  has a trigonal bipyramidal structure with  $sp^3d$ -hybridisation.



Due to presence of longer and weaker axial bonds  $\text{PCl}_5$  is a reactive molecule.

Hence, statement in option (c) is incorrect while the remaining options contain correct statements.

30. (c)  $\text{Be}(\text{OH})_2$  is amphoteric in nature as it reacts with acid and alkali both as :



This amphoteric nature of Be is due to small size of Be. The other hydroxides of alkaline earth metals are basic in nature.

31. (d) The order of energy of orbitals can be calculated from  $(n + l)$  rule. The lower the value of  $(n + l)$  for an orbital, lower is its energy. If two orbitals have same  $(n + l)$  value, the orbital with lower value of  $n$  has the lower energy.

(i)  $6p = 6 + 1 = 7$

(ii)  $5f = 5 + 3 = 8$

(iii)  $4d = 4 + 2 = 6$

(iv)  $5p = 5 + 1 = 6$

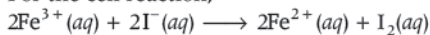
$\therefore$  The order of decreasing energy will be

$$5f > 6p > 5p > 4d.$$

32. (d) The relation between  $E_{\text{cell}}^{\circ}$  and  $\Delta_r G^{\circ}$  is as follows:

$$\Delta_r G^{\circ} = -nFE_{\text{cell}}^{\circ}$$

For the cell reaction,



$$n = 2$$

Given,  $F = 96500 \text{ C mol}^{-1}$ ,  $E_{\text{cell}}^{\circ} = 0.24 \text{ V}$

Now, we know that  $\Delta_r G^{\circ} = -nFE_{\text{cell}}^{\circ}$

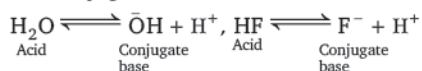
On substituting the given values in above equation we get

$$\begin{aligned}\Delta_r G^{\circ} &= -2 \text{ mol} \times 96500 \text{ C mol}^{-1} \times 0.24 \text{ J mol}^{-1} \\ &= -46320 \text{ J mol}^{-1} \\ &= -46.32 \text{ KJ mol}^{-1}\end{aligned}$$

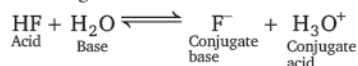
Hence, option (d) is correct.

33. (b) An acid on losing a proton produces a species which has the tendency to accept  $\text{H}^{+}$ .

It is called conjugate base of that acid.

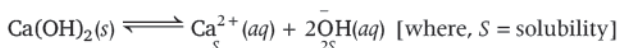


Water ( $\text{H}_2\text{O}$ ) is amphoteric in nature and thus act both as an acid and base. e.g.



34. (c) Malachite is an ore of copper and its composition is  $\text{CuCO}_3 \cdot \text{Cu(OH)}_2$ . Azurite ( $\text{Cu(OH)}_2$ ) and copper pyrites ( $\text{CuFeS}_2$ ) are also the ores of copper (Cu). Magnetite ( $\text{Fe}_3\text{O}_4$ ) is an ore of iron (Fe). Hence, option (c) is correct.

35. (d) For the reaction,



$$K_{\text{sp}} = [\text{Ca}^{2+}][\text{OH}^{-}]^2 = S(2S)^2 \quad \dots (i)$$

Given,  $\text{pH} = 9$

We know that,  $\text{pH} + \text{pOH} = 14$

$$\therefore \text{pOH} = 14 - 9 = 5$$

$$\text{pOH} = -\log[\text{OH}^{-}]$$

$$5 = -\log[\text{OH}^{-}] \quad \text{or,} \quad [\text{OH}^{-}] = 10^{-5}$$

From above equation,

$$[\text{OH}^{-}] = 2S = 10^{-5}$$

$$\therefore S = \frac{10^{-5}}{2}$$

On substituting the value of 'S' in eqn. (i), we get

$$K_{\text{sp}} = 4S^3 = 4\left(\frac{10^{-5}}{2}\right)^3 = 0.5 \times 10^{-15}$$

36. (d) All the tetrahalides of group 14 elements are covalent in nature and  $sp^3$ -hybridised with tetrahedral geometry. Exceptions are  $\text{SnF}_4$  and  $\text{PbF}_4$  which are ionic in nature. Thus, statement (d) is incorrect while the remaining statements are correct.

37. (a) **Key Idea** The colloidal particles acquire positive or negative charge by preferential adsorption of positive ions on negative ions.

When silver nitrate ( $\text{AgNO}_3$ ) solution is added to potassium iodide (KI) solution, the precipitated silver iodide ( $\text{AgI}$ ) adsorbs iodide ions ( $\text{I}^{-}$ ) from the dispersion medium and negatively charged colloidal solution results. Among the given options, in option (a) and option (b) millimole of KI is higher than  $\text{AgNO}_3$ . Hence, both the mixture will lead to the formation of negatively charged colloidal  $[\text{AgI}]\text{I}^{-}$  sol.

**Note** The most appropriate option is (a) because millimole of KI ( $50 \text{ mL} \times 2 \text{ M} = 100 \text{ mmol}$ ) is maximum in this option.

38. (d) **Key Idea** The binary liquid mixtures having the same composition in liquid and vapour phase and boil at constant temperature are called azeotropic mixtures or azeotropes.

The solutions that show large negative deviation from Raoult's law forms maximum boiling azeotrope. e.g. nitric acid and water.

The remaining option containing different mixtures forms minimum boiling azeotrope.

39. (b) We known that,

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{2.303 RT}{nF} \log Q$$

At equilibrium,  $Q = K_{\text{eq}}$  and  $E_{\text{cell}} = 0$

$$\therefore E_{\text{cell}}^{\circ} = \frac{2.303 RT}{nF} \log K_{\text{eq}} \quad \dots (i)$$

Given,  $E_{\text{cell}}^{\circ} = 0.59 \text{ V}$

$$\frac{2.303 RT}{F} = 0.059 \text{ V,}$$

$$T = 298\text{K, } n = 1$$

On substituting the given values in eqn. (i), we get

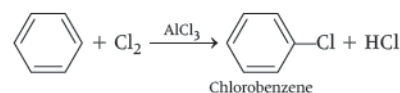
$$0.59 = \frac{0.059}{1} \log K_{\text{eq}}$$

$$\therefore \log K_{\text{eq}} = \frac{0.59}{0.059} = 10$$

$$\Rightarrow K_{\text{eq}} = 10^{10} = 1 \times 10^{10}$$

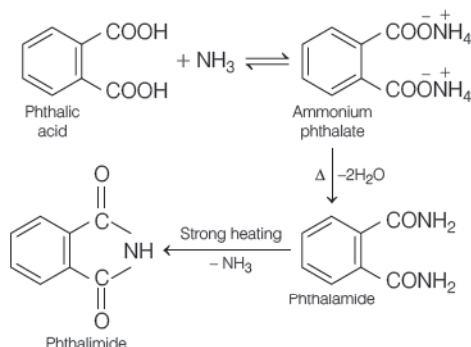
Thus, option (b) is correct.

40. (a) Halogenation of benzene is an example of electrophilic substitution reaction.



Reaction given in option (b) is an example of addition reaction, reaction in option (c) is an example of nucleophilic substitution and option (d) is an example of substitution reaction.

41. (a) Carboxylic acids react with ammonia to give ammonium salt which on further heating at high temperature give amides. Further, on strong heating, ammonia is removed from phthalamide and phthalimide is formed. The reaction takes place as follows:

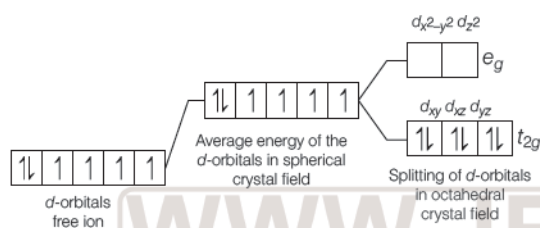


42. (a) In K<sub>4</sub>[Fe(CN)<sub>6</sub>],

$$Fe^{2+} = [Ar]3d^64s^0.$$

CN<sup>-</sup> is a strong field ligand and as it approaches the metal ion, the electrons must pair up.

The splitting of the *d*-orbitals in two sets of orbitals in an octahedral complex, K<sub>4</sub>[Fe(CN)<sub>6</sub>] may be represented as :



Hence, the electronic configuration of the central atom in K<sub>4</sub>[Fe(CN)<sub>6</sub>] is *t*<sub>2g</sub><sup>6</sup>*e*<sub>g</sub><sup>0</sup>.

43. (b) **Key Idea** For first order reaction,

$$t = \frac{2.303}{K} \log \frac{a}{a-x}$$

where, *a* = initial concentration, *a* - *x* = final concentration.

Let the initial concentration (*a*) = 100

After time *t*, final concentration

$$(a-x) = 100 - 99 = 1$$

We know that,  $t = \frac{2.303}{K} \log \frac{a}{a-x}$

On substituting the given values in above eqn. we get

$$t = \frac{2.303}{K} \log \frac{100}{1} = \frac{2.303}{K} \log 10^2 = \frac{2.303}{K} \times 2 \log 10 = \frac{4.606}{K}$$

Thus, option (b) is correct.

44. (d) **Key Idea** For an isothermal irreversible expansion,

$$W_{\text{irrev}} = -p_{\text{ext}} (V_2 - V_1)$$

Given, *V*<sub>1</sub> = 0.1 L, *V*<sub>2</sub> = 0.25 L, *p*<sub>ext</sub> = 2 bar

We know that,

$$W_{\text{irrev}} = -p_{\text{ext}} (V_2 - V_1)$$

On substituting the given values in the above equation, we get

$$\begin{aligned} W_{\text{irrev}} &= -2 \text{ bar} (0.25 - 0.1) \text{ L} \\ &= -2 \times 0.15 \text{ L bar} = -0.3 \text{ L bar} \\ &= -0.3 \times 100 \text{ J} \quad [\because 1 \text{ L bar} = 100 \text{ J}] \\ &= -30 \text{ J} \end{aligned}$$

45. (a) All enzymes that utilise ATP in phosphate transfer require magnesium as the cofactor. The cofactor are required by the enzyme for their efficient activity.

# NEET (Odisha)

## SOLVED PAPER 2019

Held on : 20 May 2019

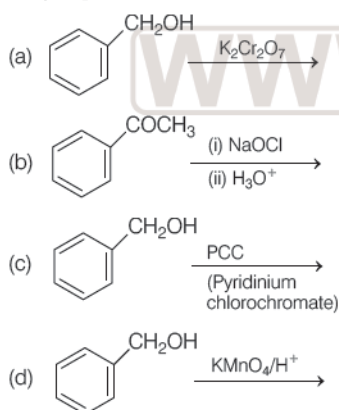
Duration : 3 Hours

Max. Marks : 180

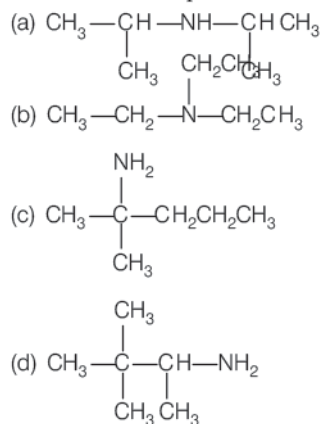
### Important Instructions

- Test Booklet contains 45 questions.
- Each question carries 4 marks. For each correct response, the candidate will get 4 marks.
- For each incorrect response, one mark will be deducted from the total scores.
- The maximum marks are 180.

1. The reaction that does not give benzoic acid as the major product is



2. The amine that reacts with Hinsberg's reagent to give an alkali insoluble product is



3. Which structure(s) of proteins remains(s) intact during denaturation process?

- (a) Both secondary and tertiary structures  
(b) Primary structure only  
(c) Secondary structure only  
(d) Tertiary structure only

4. The polymer that is used as a substitute for wool in making commercial fibres is

- (a) melamine (b) nylon-6, 6  
(c) polyacrylonitrile (d) buna-N

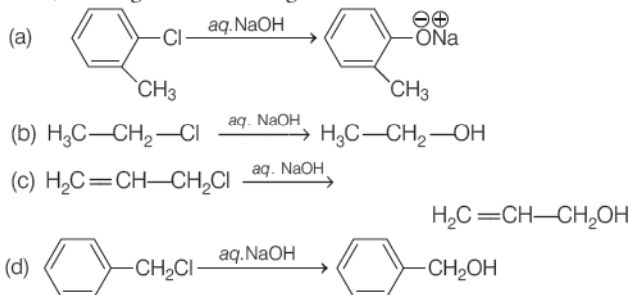
5. The artificial sweetener stable at cooking temperature and does not provide calories is

- (a) saccharin (b) aspartame  
(c) sucralose (d) alitame

6. The liquified gas that is used in dry cleaning along with a suitable detergent is

- (a) water gas (b) petroleum gas  
(c) NO<sub>2</sub> (d) CO<sub>2</sub>

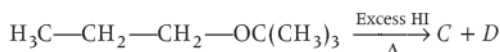
7. The hydrolysis reaction that takes place at the slowest rate, among the following is



8. When vapours of a secondary alcohol is passed over heated copper at 573 K, the product formed is

(a) a carboxylic acid (b) an aldehyde  
(c) a ketone (d) an alkene

9. The major products C and D formed in the following reactions respectively are



(a)  $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{I}$  and  $\text{I}-\text{C}(\text{CH}_3)_3$   
(b)  $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{OH}$  and  $\text{I}-\text{C}(\text{CH}_3)_3$   
(c)  $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{I}$  and  $\text{HO}-\text{C}(\text{CH}_3)_3$   
(d)  $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{OH}$  and  $\text{HO}-\text{C}(\text{CH}_3)_3$

10. Match the oxide given in column A with its property given in column B.

| Column-A                   | Column-B       |
|----------------------------|----------------|
| 1. $\text{Na}_2\text{O}$   | i. Neutral     |
| 2. $\text{Al}_2\text{O}_3$ | ii. Basic      |
| 3. $\text{N}_2\text{O}$    | iii. Acidic    |
| 4. $\text{Cl}_2\text{O}_7$ | iv. Amphoteric |

Which of the following options has all correct pairs?

1 2 3 4      1 2 3 4  
(a) (ii), (i), (iv), (iii) (b) (iii), (ii), (i), (iv)  
(c) (i), (ii), (ii), (iii) (d) (ii), (iv), (i), (iii)

11. Match the catalyst with the process

| Catalyst                                      | Process                                                                       |
|-----------------------------------------------|-------------------------------------------------------------------------------|
| 1. $\text{V}_2\text{O}_5$                     | i. The oxidation of ethyne to ethanal                                         |
| 2. $\text{TiCl}_4 + \text{Al}(\text{CH}_3)_3$ | ii. Polymerisation of alkynes                                                 |
| 3. $\text{PdCl}_2$                            | iii. Oxidation of $\text{SO}_2$ in the manufacture of $\text{H}_2\text{SO}_4$ |
| 4. Nickel complexes                           | iv. Polymerisation of ethylene                                                |

Which of the following is the correct option?

1 2 3 4      1 2 3 4  
(a) (iii), (iv), (i), (ii) (b) (i), (ii), (iii), (iv)  
(c) (i), (iii), (ii), (iv) (d) (iii), (i), (iv), (ii)

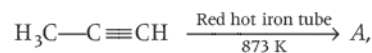
12. The most stable carbocation, among the following is

(a)  $(\text{CH}_3)_3\text{C}-\overset{\oplus}{\text{C}}\text{H}-\text{CH}_3$   
(b)  $\text{CH}_3-\text{CH}_2-\overset{\oplus}{\text{C}}\text{H}-\text{CH}_2-\text{CH}_3$   
(c)  $\text{CH}_3-\overset{\oplus}{\text{C}}\text{H}-\text{CH}_2-\text{CH}_2-\text{CH}_3$   
(d)  $\text{CH}_3-\text{CH}_2-\overset{\oplus}{\text{C}}\text{H}_2$

13. The alkane that gives only one mono-chloro product on chlorination with  $\text{Cl}_2$  in presence of diffused sunlight is

(a) 2, 2-dimethylbutane (b) neopentane  
(c) n-pentane (d) isopentane

14. In the following reaction,



the number of sigma( $\sigma$ ) bonds present in the product A is  
(a) 21 (b) 9 (c) 24 (d) 18

15. Aluminium chloride in acidified aqueous solution forms a complex 'A', in which hybridisation state of Al is 'B'. What are 'A' and 'B', respectively?

(a)  $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ ,  $sp^3d^2$  (b)  $[\text{Al}(\text{H}_2\text{O})_4]^{3+}$ ,  $sp^3$   
(c)  $[\text{Al}(\text{H}_2\text{O})_4]^{3+}$ ,  $dsp^2$  (d)  $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ ,  $d^2sp^3$

16. Which of the following compounds is used in cosmetic surgery?

(a) Silica (b) Silicates (c) Silicones (d) Zeolites

17. Identify the incorrect statement.

(a) The scientific and technological process used for isolation of the metal from its ore is known as metallurgy  
(b) Minerals are naturally occurring chemical substances in the earth's crust  
(c) Ores are minerals that may contain a metal  
(d) Gangue is an ore contaminated with undesired materials

18. A compound 'X' upon reaction with  $\text{H}_2\text{O}$  produced a colourless gas 'Y' with rotten fish smell. Gas 'Y' is absorbed in a solution of  $\text{CuSO}_4$  to give  $\text{Cu}_3\text{P}_2$  as one of the products. Predict the compound 'X'

(a)  $\text{Ca}_3\text{P}_2$  (b)  $\text{NH}_4\text{Cl}$  (c)  $\text{As}_2\text{O}_3$  (d)  $\text{Ca}_3(\text{PO}_4)_2$

19. Which of the following oxoacids of phosphorus has strongest reducing property?

(a)  $\text{H}_4\text{P}_2\text{O}_7$  (b)  $\text{H}_3\text{PO}_3$  (c)  $\text{H}_3\text{PO}_2$  (d)  $\text{H}_3\text{PO}_4$

20. Identify the correct formula of oleum from the following

(a)  $\text{H}_2\text{S}_2\text{O}_7$  (b)  $\text{H}_2\text{SO}_3$  (c)  $\text{H}_2\text{SO}_4$  (d)  $\text{H}_2\text{S}_2\text{O}_8$

21. When neutral or faintly alkaline  $\text{KMnO}_4$  is treated with potassium iodide, iodide ion is converted into 'X'. 'X' is

(a)  $\text{I}_2$  (b)  $\text{IO}_4^-$  (c)  $\text{IO}_3^-$  (d)  $\text{IO}^-$

22. The Crystal Field Stabilisation Energy (CFSE) for  $[\text{CoCl}_6]^{4-}$  is  $18000\text{ cm}^{-1}$ . The CFSE for  $[\text{CoCl}_4]^{2-}$  will be

(a)  $6000\text{ cm}^{-1}$  (b)  $16000\text{ cm}^{-1}$   
(c)  $18000\text{ cm}^{-1}$  (d)  $8000\text{ cm}^{-1}$

23. Following limiting molar conductivities are given as

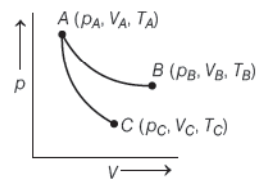
$$\lambda_m^0(\text{H}_2\text{SO}_4) = x\text{ S cm}^2\text{mol}^{-1}$$

$$\lambda_m^0(\text{K}_2\text{SO}_4) = y\text{ S cm}^2\text{mol}^{-1}$$

$$\lambda_m^0(\text{CH}_3\text{COOK}) = z\text{ S cm}^2\text{mol}^{-1}$$

$$\lambda_m^0(\text{in S cm}^2\text{mol}^{-1}) \text{ for } \text{CH}_3\text{COOH} \text{ will be}$$

(a)  $x - y + 2z$  (b)  $x + y - z$   
(c)  $x - y + z$  (d)  $\frac{(x - y)}{2} + z$

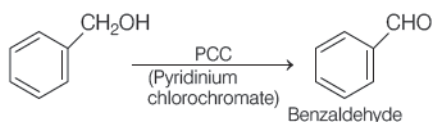
24. A first order reaction has a rate constant of  $2.303 \times 10^{-3} \text{ s}^{-1}$ . The time required for 40 g of this reactant to reduce to 10 g will be  
[Given that  $\log_{10} 2 = 0.3010$ ]  
(a) 230.3 s (b) 301 s (c) 2000 s (d) 602 s
25. For a reaction, activation energy  $E_a = 0$  and the rate constant at 200 K is  $1.6 \times 10^6 \text{ s}^{-1}$ . The rate constant at 400 K will be  
[Given that gas constant  $R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$ ]  
(a)  $3.2 \times 10^4 \text{ s}^{-1}$  (b)  $1.6 \times 10^6 \text{ s}^{-1}$   
(c)  $1.6 \times 10^3 \text{ s}^{-1}$  (d)  $3.2 \times 10^6 \text{ s}^{-1}$
26. The correct option representing a Freundlich adsorption isotherm is  
(a)  $\frac{x}{m} = kp^{0.3}$  (b)  $\frac{x}{m} = kp^{2.5}$   
(c)  $\frac{x}{m} = kp^{-0.5}$  (d)  $\frac{x}{m} = kp^{-1}$
27. Which of the following is paramagnetic?  
(a)  $\text{N}_2$  (b)  $\text{H}_2$  (c)  $\text{Li}_2$  (d)  $\text{O}_2$
28. Which of the following is the correct order of dipole moment?  
(a)  $\text{NH}_3 < \text{BF}_3 < \text{NF}_3 < \text{H}_2\text{O}$  (b)  $\text{BF}_3 < \text{NF}_3 < \text{NH}_3 < \text{H}_2\text{O}$   
(c)  $\text{BF}_3 < \text{NH}_3 < \text{NF}_3 < \text{H}_2\text{O}$  (d)  $\text{H}_2\text{O} < \text{NF}_3 < \text{NH}_3 < \text{BF}_3$
29. Crude sodium chloride obtained by crystallisation of brine solution does not contain  
(a)  $\text{MgSO}_4$  (b)  $\text{Na}_2\text{SO}_4$  (c)  $\text{MgCl}_2$  (d)  $\text{CaSO}_4$
30. Which of the alkali metal chloride ( $\text{MCl}$ ) forms its dihydrate salt ( $\text{MCl} \cdot 2\text{H}_2\text{O}$ ) easily?  
(a)  $\text{LiCl}$  (b)  $\text{CsCl}$  (c)  $\text{RbCl}$  (d)  $\text{KCl}$
31. The pH of 0.01 M  $\text{NaOH}$  (aq) solution will be  
(a) 7.01 (b) 2 (c) 12 (d) 9
32. Which of the following cannot act both as Bronsted acid and as Bronsted base?  
(a)  $\text{HCO}_3^-$  (b)  $\text{NH}_3$  (c)  $\text{HCl}$  (d)  $\text{HSO}_4^-$
33. The molar solubility of  $\text{CaF}_2$  ( $K_{sp} = 5.3 \times 10^{-11}$ ) in 0.1 M solution of  $\text{NaF}$  will be  
(a)  $5.3 \times 10^{11} \text{ mol L}^{-1}$  (b)  $5.3 \times 10^{-8} \text{ mol L}^{-1}$   
(c)  $5.3 \times 10^{-9} \text{ mol L}^{-1}$  (d)  $5.3 \times 10^{-10} \text{ mol L}^{-1}$
34. The oxidation state of Cr in  $\text{CrO}_5$  is  
(a) -6 (b) +12 (c) +6 (d) +4
35. The number of hydrogen bonded water molecule(s) associated with  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$  is  
(a) 3 (b) 1 (c) 2 (d) 5
36. Formula of nickel oxide with metal deficiency defect in its crystal is  $\text{Ni}_{0.98}\text{O}$ . The crystal contains  $\text{Ni}^{2+}$  and  $\text{Ni}^{3+}$  ions. The fraction of nickel existing as  $\text{Ni}^{2+}$  ions in the crystal is  
(a) 0.96 (b) 0.04 (c) 0.50 (d) 0.31
37. Which of the following statements is correct regarding a solution of two compounds A and B exhibiting positive deviation from ideal behaviour?  
(a) Intermolecular attractive forces between A—A and B—B are stronger than those between A—B.  
(b)  $\Delta_{\text{mix}} H = 0$  at constant  $T$  and  $p$   
(c)  $\Delta_{\text{mix}} V = 0$  at constant  $T$  and  $p$   
(d) Intermolecular attractive forces between A—A and B—B are equal to those between A—B.
38. In water saturated air the mole fraction of water vapour is 0.02. If the total pressure of the saturated air is 1.2 atm, the partial pressure of dry air is  
(a) 1.18 atm (b) 1.76 atm (c) 1.176 atm (d) 0.98 atm
39. The standard electrode potential ( $E^\circ$ ) values of  $\text{Al}^{3+} / \text{Al}$ ,  $\text{Ag}^+ / \text{Ag}$ ,  $\text{K}^+ / \text{K}$  and  $\text{Cr}^{3+} / \text{Cr}$  are -1.66 V, 0.80 V, 2.93 V and -0.74 V, respectively. The correct decreasing order of reducing power of the metal is  
(a)  $\text{Ag} > \text{Cr} > \text{Al} > \text{K}$  (b)  $\text{K} > \text{Al} > \text{Cr} > \text{Ag}$   
(c)  $\text{K} > \text{Al} > \text{Ag} > \text{Cr}$  (d)  $\text{Al} > \text{K} > \text{Ag} > \text{Cr}$
40. The density of 2 M aqueous solution of  $\text{NaOH}$  is 1.28  $\text{g cm}^{-3}$ . The molality of the solution is [Given that molecular mass of  $\text{NaOH} = 40 \text{ g mol}^{-1}$ ]  
(a) 1.20 m (b) 1.56 m (c) 1.67 m (d) 1.32 m
41. Orbital having 3 angular nodes and 3 total nodes is  
(a) 5 p (b) 3 d (c) 4 f (d) 6 d
42. In hydrogen atom, the de-Broglie wavelength of an electron in the second Bohr orbit is  
[Given that, Bohr radius,  $a_0 = 52.9 \text{ pm}$ ]  
(a) 211.6 pm (b)  $211.6 \pi \text{ pm}$   
(c)  $52.9 \pi \text{ pm}$  (d) 105.8 pm
43. The volume occupied by 1.8 g of water vapour at  $374^\circ\text{C}$  and 1 bar pressure will be  
[Use  $R = 0.083 \text{ bar L K}^{-1} \text{ mol}^{-1}$ ]  
(a) 96.66 L (b) 55.87 L (c) 3.10 L (d) 5.37 L
44. An ideal gas expands isothermally from  $10^{-3} \text{ m}^3$  to  $10^{-2} \text{ m}^3$  at 300 K against a constant pressure of  $10^5 \text{ Nm}^{-2}$ . The work done on the gas is  
(a) +270 kJ (b) -900 J (c) +900 kJ (d) -900 kJ
45. Reversible expansion of an ideal gas under isothermal and adiabatic conditions are as shown in the figure.  
  
 $AB \rightarrow$  Isothermal expansion  
 $AC \rightarrow$  Adiabatic expansion  
Which of the following option is not correct?  
(a)  $\Delta S_{\text{isothermal}} > \Delta S_{\text{adiabatic}}$  (b)  $T_A = T_B$   
(c)  $W_{\text{isothermal}} > W_{\text{adiabatic}}$  (d)  $T_C > T_A$

## Answers

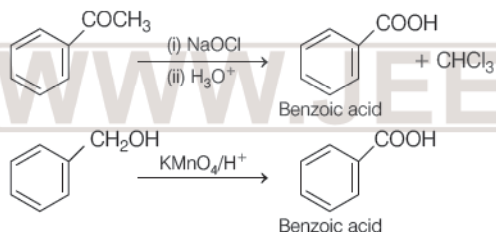
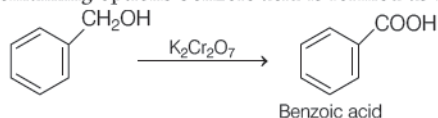
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|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 1.  | (c) | 2.  | (a) | 3.  | (b) | 4.  | (c) | 5.  | (c) | 6.  | (d) | 7.  | (a) | 8.  | (c) | 9.  | (a) | 10. | (d) |
| 11. | (a) | 12. | (c) | 13. | (b) | 14. | (a) | 15. | (a) | 16. | (c) | 17. | (d) | 18. | (a) | 19. | (c) | 20. | (a) |
| 21. | (c) | 22. | (d) | 23. | (d) | 24. | (d) | 25. | (b) | 26. | (a) | 27. | (d) | 28. | (b) | 29. | (a) | 30. | (a) |
| 31. | (c) | 32. | (c) | 33. | (c) | 34. | (c) | 35. | (b) | 36. | (a) | 37. | (a) | 38. | (c) | 39. | (b) | 40. | (c) |
| 41. | (c) | 42. | (b) | 43. | (d) | 44. | (b) | 45. | (d) |     |     |     |     |     |     |     |     |     |     |

## Answer with Explanations

1. (c) Primary aromatic alcohols on reaction with pyridinium chlorochromate (PCC) which is a mild oxidising agent forms aromatic aldehydes.



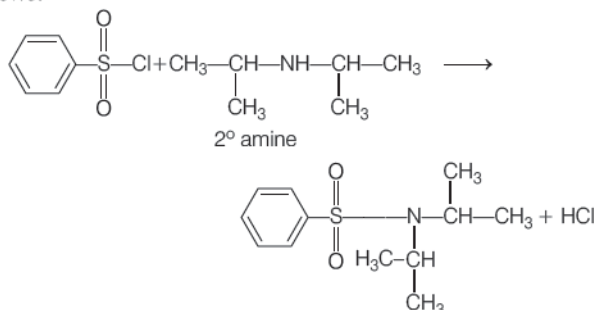
In the remaining options benzoic acid is formed as follows:



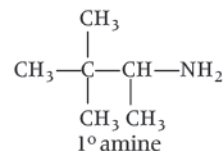
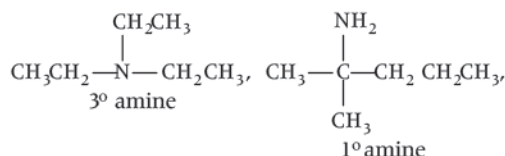
Thus, option (c) is correct.

2. (a) Secondary amines on reaction with Hinsberg's reagent (benzene sulphonyl chloride) forms N,N-dialkyl benzene sulphonamide. The product formed is not acidic as no H-atom is attached to N thus it is insoluble in alkali.

The reaction of 2° amine given in option (a) takes place as follows:



On the other hand, 1° amines react with Hinsberg's reagent to yield N-alkyl benzene sulphonamide which is soluble in alkali and 3° amines do not react with  $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$ . The remaining options contain 1° and 3° amines.

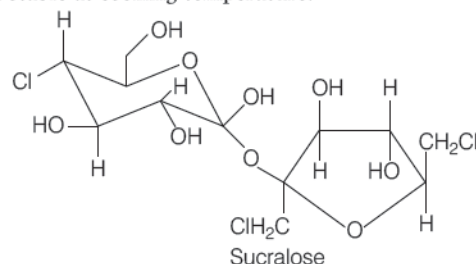


Thus, option (a) is correct.

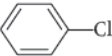
3. (b) The loss in biological activity of proteins due to unfolding of globules and uncoiling of helix is called denaturation of proteins. During the denaturation, primary structure remains intact but secondary and tertiary structures are destroyed. e.g. The coagulation of egg white on boiling, curdling of milk caused by bacteria due to formation of lactic acid.

4. (c) Polyacrylonitrile or orlon or acrilan,  $\text{-(CH}_2\text{-CH(CN))}_n\text{-}$  is an addition homopolymer of monomer  $\text{CH}_2=\text{CHCN}$  (vinyl cyanide). It is used in making synthetic fibres and synthetic wool. Thus, it is a substitute for wool in making commercial fibres.

5. (e) Sucralose is a trichloro derivative of sucrose and is about 650 times sweeter than cane sugar. It is a zero calorie sugar and stable at cooking temperature.

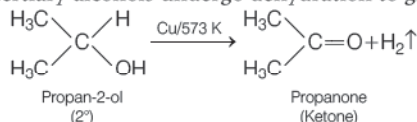


6. (d) Solvents used to dry clean clothes are usually chlorinated compounds which are carcinogenic. Suitable detergents which work in liquid carbon dioxide have been discovered to replace the chlorinated compounds. Thus,  $\text{CO}_2$  is the liquified gas that is used in dry cleaning along with a suitable detergent.

7. (a)  is a benzylic halide. Thus, there is a partial

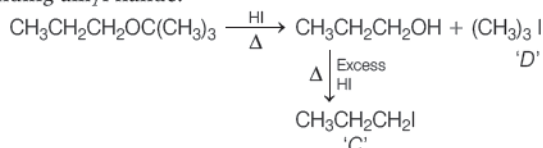
double bond character between  $sp^3$ -hybridised C atom next to an aromatic ring and Cl. It is most difficult to break this bond and hence it undergoes hydrolysis reaction with slowest rate.

8. (c) When vapours of alcohols are passed over heated copper at 573 K, primary and secondary alcohols undergo dehydrogenation to give aldehydes and ketones, respectively. While tertiary alcohols undergo dehydration to give alkenes.



Thus, option (c) is correct.

9. (a) Ethers are readily cleaved by heating in presence of halogen acids to form alcohol and an alkyl halide. In case of unsymmetrical ethers, halogen goes preferentially with smaller alkyl group or more stable carbocation. If excess of acid is used then only alkyl halide is formed because alcohol formed reacts further with halogen acid to form corresponding alkyl halide.



10. (d) **Key Idea** Metal oxides are basic, non-metal oxides are acidic while semi-metal oxides are amphoteric in nature. Thus, the basic character of oxides decreases across the period and increases down the group.

The correct match of oxide with its property are as follows:

$\text{Na}_2\text{O}$ - Basic                       $\text{Al}_2\text{O}_3$ - Amphoteric

$\text{N}_2\text{O}$ - Neutral                       $\text{Cl}_2\text{O}_7$ - Acidic

Thus, option (d) is correct.

11. (a) (1)  $\text{V}_2\text{O}_5$  - (iii) Oxidation of  $\text{SO}_2$  in the manufacture of  $\text{H}_2\text{SO}_4$  by contact process.

(2)  $\text{TiCl}_4 + \text{Al}(\text{CH}_3)_3$  - (Ziegler-Natta catalyst) - (iv) Polymerisation of ethylene.

(3)  $\text{PdCl}_2$  - (i) Oxidation of ethyne to ethanal.

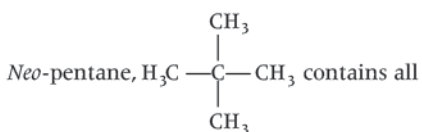
(4) Nickel complexes - (ii) Polymerisation of alkynes.

Thus, the correct match is

(1)  $\rightarrow$  (iii), (2)  $\rightarrow$  (iv), (3)  $\rightarrow$  (i), (4)  $\rightarrow$  (ii)

12. (c)  $\text{CH}_3\text{CH}^+\text{CH}_2\text{CH}_2\text{CH}_3$  is the most stable carbocation among the given carbocations. It is because the number of  $\alpha$ -H atom attached to carbocation is maximum in  $\text{CH}_3\text{CH}^+\text{CH}_2\text{CH}_2\text{CH}_3$ . Thus, it has maximum number of hyperconjugating structure hence, it is most stable.

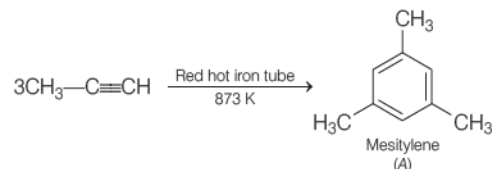
13. (b) **Key Idea** Alkanes which contain all equivalent hydrogen atoms forms only one monochloro derivative on halogenation.



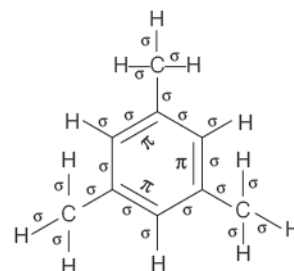
equivalent hydrogen atoms. So, it will give only one monochloro derivative on halogenation.

14. (a) **Key Idea** Single bond =  $1\sigma$ , Double bond =  $1\sigma, 1\pi$ ; Triple bond =  $1\sigma$  and  $2\pi$ .

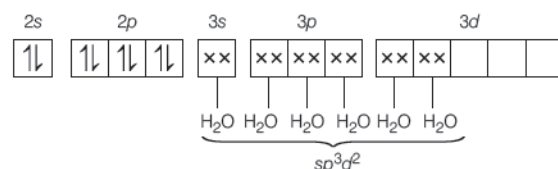
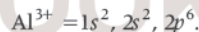
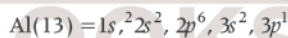
Alkynes form aromatic compounds when their vapours are passed over red hot copper or iron tubes.



The number of sigma ( $\sigma$ ) bonds present in the product (A) is 21.



15. (a) Aluminium chloride in acidified aqueous solution forms an octahedral complex. Aqueous solution is mostly water and when the compound is dissolved in acidified aqueous solution, the water fills its vacancies and an octahedral complex, 'A' which is  $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$  is formed.

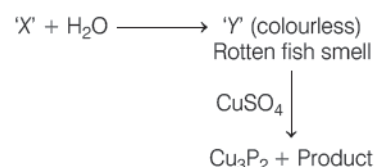


Thus, the hybridisation state of Al in  $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$  is  $sp^3d^2$  (B).

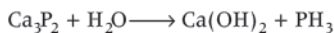
16. (c) Silicones are synthetic organo-silicon polymers containing repeated  $\text{R}_2\text{SiO}$  units. It is used in surgical and cosmetic implants. They are also used as low temperature lubricants, in making waterproof clothes and paper, etc.

17. (d) The earthly impurities like sand, clay, mica, etc., associated with ores are called gangue or matrix. In other words, contaminated undesired materials present in an ore is called gangue. Thus, statement (d) is incorrect while other options contain correct statements.

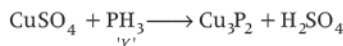
18. (a) The given road map problem is



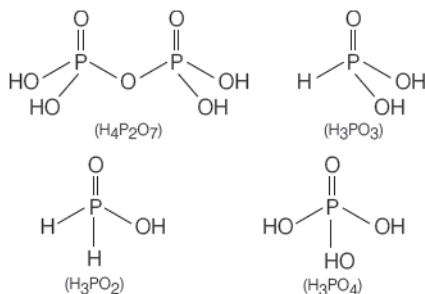
As compound 'X' produces a colorless gas 'Y' with rotten fish smell, it can be  $\text{Ca}_3\text{P}_2$  (calcium phosphide). The reaction takes place as follows :



Colorless gas with rotten fish smell. 'Y'



19. (c) P—H bonds found in phosphorus acids have reducing properties. Thus, reducing property is directly proportional to number of P—H bonds. The structures of given oxoacids of phosphorus are as follows:



$\text{H}_3\text{PO}_2$  contains 2P—H bonds which is maximum among given options. Thus,  $\text{H}_3\text{PO}_2$  has strongest reducing property.

20. (a) The correct formula of oleum is  $\text{H}_2\text{S}_2\text{O}_7$  which is also known as pyrosulphuric acid.



21. (c) When neutral or faintly alkaline  $\text{KMnO}_4$  is treated with potassium iodide,  $\text{KMnO}_4$  is converted into  $\text{MnO}_2$  while iodide ( $\text{I}^-$ ) ion is converted into  $\text{IO}_3^-$  (iodate). The reaction takes place as follows:



22. (d)  $[\text{CoCl}_6]^{4-}$  is an octahedral while  $[\text{CoCl}_4]^{2-}$  is a tetrahedral complex.

$$\Delta_{\text{octahedral}} = 18000 \text{ cm}^{-1}$$

We know that,

$$\begin{aligned} \Delta_{\text{tetrahedral}} &= \frac{4}{9} \Delta_{\text{octahedral}} \\ &= \frac{4}{9} \times 18000 \text{ cm}^{-1} = 8000 \text{ cm}^{-1} \end{aligned}$$

Hence, option (d) is correct.

23. (d) **Key Idea** According to Kohlrausch's law,

$$\lambda_m^\circ \text{ for } A_x B_y = x \lambda_{(A)}^{y+} + y \lambda_{(B)}^{x-}$$

where,  $\lambda_m^\circ$  = limiting molar conductivity of electrolyte and  $\lambda_{(A)}^{y+}$  and  $\lambda_{(B)}^{x-}$  are the limiting molar conductivities of cation ( $A^{y+}$ ) and anion ( $B^{x-}$ ), respectively.

$$\text{Given } \lambda_{\text{m}(\text{H}_2\text{SO}_4)}^\circ = 2\lambda_{\text{H}^+}^\circ + \lambda_{\text{SO}_4^{2-}}^\circ = x \text{ S cm}^2 \text{ mol}^{-1} \dots (i)$$

$$\lambda_{\text{m}(\text{K}_2\text{SO}_4)}^\circ = 2\lambda_{\text{K}^+}^\circ + \lambda_{\text{SO}_4^{2-}}^\circ = y \text{ S cm}^2 \text{ mol}^{-1} \dots (ii)$$

$$\lambda_{\text{m}(\text{CH}_3\text{COOK})}^\circ = \lambda_{\text{CH}_3\text{COO}^-}^\circ + \lambda_{\text{K}^+}^\circ = z \text{ S cm}^2 \text{ mol}^{-1}$$

...(iii)

$$\text{To find : } \lambda_{\text{m}(\text{CH}_3\text{COOH})}^\circ = \lambda_{\text{CH}_3\text{COO}^-}^\circ + \lambda_{\text{H}^+}^\circ$$

The above equation can be obtained by

$$\text{Eq. (iii)} + \frac{\text{Eq. (i)}}{2} - \frac{\text{Eq. (ii)}}{2}$$

$$\begin{aligned} \therefore \lambda_{\text{m}(\text{CH}_3\text{COOH})}^\circ &= \lambda_{\text{m}(\text{CH}_3\text{COOK})}^\circ + \frac{\lambda_{\text{m}(\text{H}_2\text{SO}_4)}^\circ}{2} - \frac{\lambda_{\text{m}(\text{K}_2\text{SO}_4)}^\circ}{2} \\ &= \left( z + \frac{x}{2} - \frac{y}{2} \right) \text{ S cm}^2 \text{ mol}^{-1} \\ &= \left( \frac{x-y}{2} + z \right) \text{ S cm}^2 \text{ mol}^{-1} \end{aligned}$$

24. (d) For first order reaction,

$$t = \frac{2.303}{k} \log \frac{a}{a-x}$$

...(i)

Given :  $k = 2.303 \times 10^{-3} \text{ s}^{-1}$ ,  $a = 40 \text{ g}$ ,  $a-x = 10 \text{ g}$

On substituting the given values in Eq. (i), we get

$$\begin{aligned} t &= \frac{2.303}{2.303 \times 10^{-3}} \log \frac{40}{10} \\ &= 10^3 \log 2^2 = 2 \times 10^3 \times \log 2 \\ &= 2 \times 10^3 \times 0.3010 = 602 \text{ s} \end{aligned}$$

#### Alternative method

For first order reaction,

$$\begin{aligned} t_{1/2} &= \frac{0.693}{k} \\ t_{1/2}(t_{50\%}) &= \frac{0.693}{2.303 \times 10^{-3}} = 301 \text{ s} \end{aligned}$$

Also,  $t_{75\%} = 2t_{50\%}$

$$\therefore t_{75\%} = 2 \times 301 = 602 \text{ s}$$

25. (b) **Key Idea** Rate constants at two different temperatures is given by Arrhenius equation as follows:

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[ \frac{1}{T_1} - \frac{1}{T_2} \right], T_2 > T_1$$

Given,  $E_a = 0$ ,  $T_1 = 200 \text{ K}$ ,  $k_1 = 1.6 \times 10^6 \text{ s}^{-1}$

$$T_2 = 400 \text{ K},$$

$$R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$$

According to Arrhenius equation,

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[ \frac{1}{T_1} - \frac{1}{T_2} \right]$$

On substituting the given values in above equation, we get

$$\log \frac{k_2}{1.6 \times 10^6} = \frac{0}{2.303 \times 8.314} \left[ \frac{1}{200} - \frac{1}{400} \right]$$

$$\log \left( \frac{k_2}{1.6 \times 10^6} \right) = 0$$

$$\frac{k_2}{1.6 \times 10^6} = 10^0 = 1$$

$$k_2 = 1.6 \times 10^6 \text{ s}^{-1} \text{ at } 400 \text{ K}$$

Thus, option (b) is correct.

26. (a) According to Freundlich adsorption isotherm,

$$\frac{x}{m} = kp^{1/n}$$

where,  $\frac{x}{m}$  = amount of the gas adsorbed per unit mass of adsorbent

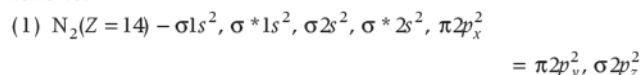
$p$  = pressure  $k$  and  $n$  = constants.

The value of  $n$  lies in between 0 to 1.

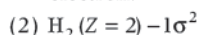
Thus,  $\frac{x}{m} = kp^{0.3}$  and option (a) is correct.

27. (d) **Key Idea** If all the electron in a molecule are paired, the molecule is diamagnetic and if there are unpaired electrons in a molecule, it is paramagnetic.

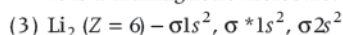
Molecular orbital configuration of given molecules are as follows:



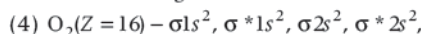
It is a diamagnetic molecule due to absence of unpaired electron.



It is a diamagnetic molecule.



It is a diamagnetic molecule.



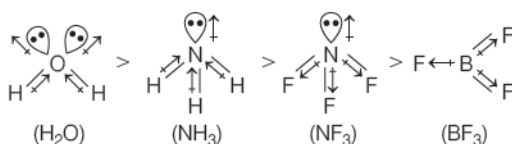
It is a paramagnetic molecule due to presence of unpaired electrons in each orbital of degenerate levels.

Thus, option (d) is correct.

28. (b)  $\text{BF}_3$  has zero dipole moment as it is symmetrical in nature.

$\text{H}_2\text{O}$  has maximum dipole moment as it possess two lone pair of electrons. Between  $\text{NH}_3$  and  $\text{NF}_3$ ,  $\text{NH}_3$  has greater dipole moment though in  $\text{NH}_3$  and  $\text{NF}_3$ , both N possesses one lone pair of electrons.

This is because in case of  $\text{NH}_3$ , the net N—H bond dipole is in the same direction as the direction of dipole of lone pair. But in case of  $\text{NF}_3$ , the direction of net dipole moment of three —N—F bonds is opposite to that of the dipole moment of the lone pair. Thus, the correct of dipole moment is



29. (a) Crude sodium chloride obtained by crystallisation of brine solution contains  $\text{CaSO}_4$ ,  $\text{Na}_2\text{SO}_4$ ,  $\text{CaCl}_2$ ,  $\text{MgCl}_2$  as impurities.  $\text{CaCl}_2$  and  $\text{MgCl}_2$  are deliquescent, i.e. absorb moisture from atmosphere and give wet appearance to the salt. Crude salt is dissolved in water and insoluble impurities are removed by filtration. Hence,  $\text{MgSO}_4$  is not present in crude sodium chloride.

30. (a) The size of the ion and its charge determines extent of hydration. Greater the charge and smaller the size of ion, greater is the extent of hydration and hence greater is the hydration enthalpy. Thus,  $\text{Li}^+$  having maximum hydration tendency forms dihydrate salt ( $\text{LiCl} \cdot 2\text{H}_2\text{O}$ ) easily.

31. (c)  $\text{NaOH}$  is a strong base, thus

$$[\text{OH}^-] = 0.01\text{M} = 10^{-2}\text{M}$$

$$\text{pOH} = -\log [\text{OH}^-] = -\log(10^{-2}) = 2$$

We know that,  $\text{pH} + \text{pOH} = 14$

$$\therefore \text{pH} = 14 - 2 = 12$$

Thus, option (c) is correct.

32. (c) **Key Idea** Bronsted acid is a substance which has a tendency to donate proton. Bronsted base is a substance which has a tendency to accept proton.

$\text{HCl}$  can act as Bronsted acid because it can only donate proton.

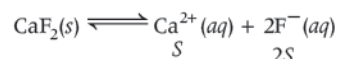


The remaining options contains substances which act both as Bronsted acid and Bronsted base.



Thus, option (c) is correct.

33. (c) Let the solubility of  $\text{CaF}_2$  in 0.1 M  $\text{NaF}$  is ' $S$ ' mol  $\text{L}^{-1}$



$$[\text{F}^-] = 2S + 0.1$$

$$K_{sp} \text{ of } \text{CaF}_2 = [\text{Ca}^{2+}][\text{F}^-]^2$$

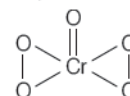
$$= [S][2S + 0.1]^2$$

$$= 53 \times 10^{-11} = [S][2S + 0.1]^2$$

$$\Rightarrow 53 \times 10^{-11} = [S][0.1]^2 \quad [\because 2S \ll 0.1]$$

$$[S] = \frac{53 \times 10^{-11}}{(0.1)^2} = 5.3 \times 10^{-9} \text{ mol L}^{-1}$$

34. (c) The structure of  $\text{CrO}_5$  is



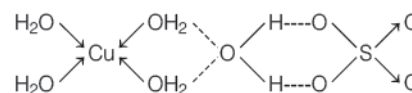
Oxidation state of Cr is +6 due to the presence of two peroxide linkages, which can be calculated as.

In  $\text{CrO}(\text{O}_2)_2$ , let the oxidation state of Cr be  $x$

$$x + (-1)4 + (-2) = 0$$

$$x - 6 = 0 \Rightarrow x = +6$$

35. (b)  $\text{Cu}^{2+}$  ion has 17 electrons in its outermost orbital.  $4\text{H}_2\text{O}$  molecules donate 4 pairs of electrons for coordinate covalent sharing. The one  $\text{H}_2\text{O}$  molecule is attached to the sulphate ion by hydrogen bonding, the oxygen being oriented to the hydrated cupric ion.



Thus, the number of hydrogen bonded water molecule associated with  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$  is 1.

36. (a) Let, in the given crystal
- $\text{Ni}_{0.98}\text{O}$

$$\text{Ni}^{2+} = x \text{ and } \text{Ni}^{3+} = 0.98 - x$$

$$\begin{aligned} \text{Total charge on } M^{2+} \text{ and } M^{3+} &= (+2)x + (+3)(0.98 - x) \\ &= 2x + 2.94 - 3x = 2.94 - x \end{aligned}$$

As metal oxide is neutral. Therefore, total charge on cations = total charge on anions.

$$2.94 - x = 2 \quad [\because \text{Charge of oxygen atom} = -2]$$

$$x = 2.94 - 2 = 0.94$$

$$\text{So, the fraction of } \text{Ni}^{2+} \text{ ions in the crystal} = \frac{0.94}{0.98} = 0.96$$

37. (a) Solution exhibits positive deviation from ideal behaviour if the intermolecular interactions, i.e., A—B interactions are more stronger than in pure components, i.e. between A—B or B—B. Thus, option (a) is correct. The remaining options are valid only for ideal solutions.

38. (c) Partial pressure of dry air = total pressure
- $\times$
- mole fraction of dry air

$$\Rightarrow p_{\text{dry air}} = p_{\text{total}} \times \lambda_{\text{dry air}}$$

$$\text{Given, } \lambda_{\text{saturated air}} = 0.02$$

$$\lambda_{\text{dry air}} = 1 - 0.02 = 0.98$$

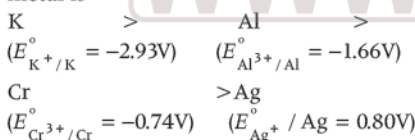
$$p_{\text{total}} = 1.2 \text{ atm}$$

$$\therefore p_{\text{dry air}} = 1.2 \text{ atm} \times 0.98 = 1.176 \text{ atm}$$

39. (b) More negative the value of standard reduction potential, higher is the reduction power.

$$\text{i.e. Reducing power} \propto \frac{1}{\text{standard reduction potential}}$$

Thus, the correct decreasing order of reducing power of the metal is



40. (c) Given, molarity = 2M [i.e. 2 mole NaOH in 1 L solution]

$$\text{Density (d)} = 1.28 \text{ cm}^{-3}$$

$$\text{Molecular weight of NaOH} = 40 \text{ g mol}^{-1}$$

We know that,

$$\text{Density} = \frac{\text{mass of solution}}{\text{volume of solution}}$$

$$\therefore \text{Mass of solution} = 1.28 \text{ g cm}^{-3} \times 1000 \text{ mL} = 1280 \text{ g}$$

Moreover, molarity

$$= \frac{\text{number of moles of solute}}{\text{volume of solution (in mL)}} \times 1000$$

$$\therefore 2 = \frac{\text{mass of solute}}{40 \times 1000} \times 1000 = 80 \text{ g}$$

$$\text{Now, mass of solution} = \text{mass of solvent} + \text{mass of solute}$$

$$1280 \text{ g} = \text{mass of solvent} + 80$$

$$\therefore \text{Mass of solvent} = 1280 - 80 = 1200 \text{ g} = 1.2 \text{ kg}$$

$$\begin{aligned} \text{Now, molality} &= \frac{\text{number of moles of solute}}{\text{mass of solvent (in kg)}} = \frac{2}{1.2} = \frac{20}{12} = \frac{5}{3} \\ &= 1.67 \text{ m} \end{aligned}$$

**Alternative method**

$$\text{Molality (m)} = \frac{M}{1000d - Mwt} \times 1000$$

where,  $M$  = molarity,  $d$  = density of solution  
 $Mwt$  = molar mass of solute.

On substituting the given values, we get

$$\begin{aligned} \text{Molality} &= \frac{2}{1000 \times 1.28 - 40} \times 1000 \\ &= \frac{2}{1280 - 40} \times 1000 = \frac{2000}{1240} = 1.612 \text{ m} \end{aligned}$$

41. (c) Angular node (
- $l$
- ) = 3

Total node = radial node + angular node

$$3 = (n - l - 1) + l$$

$$3 = n - 1 \Rightarrow n = 4$$

$\therefore$  Orbital having 3 angular nodes and 3 total nodes is  $n = 4$  [ $\because l = 3$  for  $f$ -orbital]

42. (b) According to Bohr,

$$mvr = \frac{nh}{2\pi}$$

$$2\pi r = \frac{nh}{mv} = n\lambda \quad \dots(i) \left[ \because \lambda = \frac{h}{mv} \right]$$

where,  $r$  = radius,  $\lambda$  = wavelength

$n$  = number of orbit

$$\text{Also, } r = \frac{a_0 n^2}{Z} \quad \dots(ii)$$

where,  $a_0$  = Bohr radius = 52.9 pm

$Z$  = atomic number

On substituting the value of ' $r$ ' from Eq. (ii) to Eq. (i), we get

$$\begin{aligned} n\lambda &= \frac{2\pi n^2 a_0}{Z} \Rightarrow \lambda = \frac{2\pi n a_0}{Z} \\ \lambda &= 2\pi \times 2 \times 52.9 = 211.6\pi \text{ pm} \quad [\because n = 2, Z = 1] \end{aligned}$$

43. (d) According to ideal gas equation,

$$pV = nRT$$

$$\text{or } V = \frac{nRT}{p} = \frac{w}{M.wt} \frac{RT}{p} \quad \dots(i) \left[ \because n = \frac{w}{M.wt} \right]$$

$$\text{Given, } w = 1.8 \text{ g, } T = 374^\circ \text{C}$$

$$= (374 + 273) \text{ K} = 647 \text{ K}$$

$$p = 1 \text{ bar, } R = 0.083 \text{ bar L K}^{-1} \text{ mol}^{-1}$$

On substituting the given values in Eq. (i), we get

$$V = \frac{1.8 \text{ g}}{18 \text{ g mol}^{-1}} \times \frac{0.083 \text{ bar L K}^{-1} \text{ mol}^{-1} \times 647 \text{ K}}{1 \text{ bar}} = 5.37 \text{ L}$$

44. (b) For an isothermal irreversible expansion,

$$\text{Work done (W)} = -p_{\text{ext}}(V_2 - V_1)$$

where,  $V_1$  = initial volume,  $V_2$  = final volume

$$\text{Given, } p_{\text{ext}} = 10^5 \text{ Nm}^{-2},$$

$$V_1 = 10^{-3} \text{ m}^3, V_2 = 10^{-2} \text{ m}^3$$

On substituting the given values in Eq. (i), we get,

$$\begin{aligned} W &= -10^5 \text{ Nm}^{-2} (10^{-2} \text{ m}^3 - 10^{-3} \text{ m}^3) \\ &= -10^5 \text{ Nm}^{-2} \times 10^{-3} (10 - 1) \text{ m}^3 = -900 \text{ Nm} = -900 \text{ J} \end{aligned}$$

45. (d) From first law of thermodynamics,
- $\Delta U = q + W$

In adiabatic expansion,  $q = 0$

$$\therefore \Delta U = W$$

During expansion of a gas  $w$  is negative i.e.  $w < 0$  or  $\Delta U < 0$ .

We know that,  $\Delta U = nC_V \Delta T$

$$\therefore nC_V \Delta T < 0 \text{ or } \Delta T < 0$$

$$\therefore T_C - T_A < 0 \text{ or } T_C < T_A$$

Thus, option (d) is incorrect while the remaining options are correct.